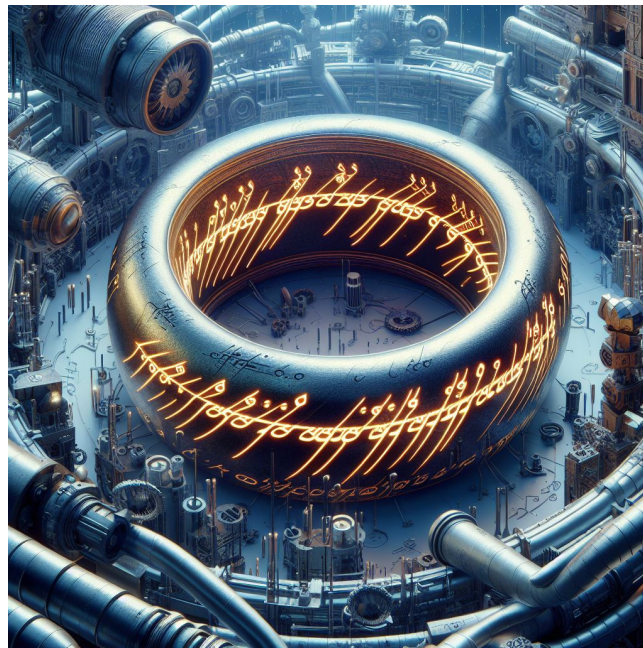


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Preliminary Design of a Bipropellant Semi-Cryogenic Blowdown Liquid Rocket Engine



The Fellowship Of The O-Ring

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Preliminary Design of a Bipropellant Semi-Cryogenic Blowdown
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Abstract

I'm Winston Wolfe. I solve problems.

Mr. Wolfe in Pulp Fiction, 1994

This report presents a detailed analysis of a proposed design for a Liquid Rocket Engine fueled by a combination of liquid oxygen and RP-1, utilizing a blowdown architecture to perform propellant delivery and define the mixture before the combustion. The investigation focuses on assessing the performance degradation associated with the nominal configuration, particularly evaluating the time-varying profiles of thrust and specific impulse resulting from pressure lowering inherent in the blowdown system. The propulsion system performance is further examined by exploring the application of additive manufacturing technology. This innovative approach enables the fabrication of a complex designed thrust chamber, leading to reductions in both mass and cost, as well as introducing uncertainties that will be discussed. Additionally, this study highlights the potential of utilizing different cooling techniques. It encompasses both traditional cooling jackets along the combustion chamber and innovative approaches such as dumb cooling, film cooling, and the integration of advanced ceramic materials like Zirconia Yttria. Moreover, a proposal for an insulating coating is presented to counteract the potential issues related to the critical liquid oxygen thermal properties.

Nomenclature

α	Divergent half- angle [deg]
β	Convergent half-angle [deg]
ΔP	Pressure losses [Pa]
δ	Injection angle [deg]
$\dot{m}$	Mass flow rate [kg/s]
ϵ	Expansion ratio $[-]$
ϵ_c	Contraction ratio $[-]$
γ	Specific heat ratio $[-]$
λ	Divergence loss coefficient $[-]$
μ	Dynamic viscosity [$Pa\cdot s$]
ϕ	Diameter [m]
ρ	Density [kg/m^3]
θ_e	Final Parabola angle [deg]
θ_i	Initial parabola angle [deg]
ξ	Impinging angle [deg]
c^*	Characteristic velocity [m/s]
C_d	Discharge coefficient $[-]$
C_N	Correction factor of nozzle $[-]$
c_p	Specific heat capacity at constant pressure [kJ/kgK]
c_T	Thrust coefficient $[-]$
$d_{channel}$	Dimension of the cooling channels [m]
f	Darcy friction factor $[-]$
g_0	Gravity constant [m/s^2]
h_{cg}	Convective heat coefficient of combusted gas [W/Km^2]
I_s	Specific impulse [s]
I_v	Volumetric specific impulse [$kg\cdot s/m^3$]
I_{tot}	Total impulse [$N\cdot s$]
k	Conductivity [W/mK]
L	Length [m]
L^*	Characteristic length of combustion chamber [m]
L_{pipes}	Length of the pipes [m]

M	Mach number $[-]$
M_{mol}	Molar mass $[g/mol]$
N	Number injectors $[-]$
N_{ch}	Number of cooling channels $[-]$
Nu	Nusselt number $[-]$
O/F	Oxidizer to fuel ratio $[-]$
P	Pressure $[Pa]$
Pr	Prandtl number $[-]$
Q	Heat flux $[W]$
q	Heat flux per unit area $[W/m^2]$
R	Universal gas constant $[KJ/kgmol]$
r	Recovery factor $[-]$
r_s	Spacecraft Radius
R_t	Thermal Radius $[ft]$
R_{ext}	Tank external radius $[m]$
R_{int}	Tank internal radius $[m]$
Ra	Roughness $[\mu m]$
Ra_{real}	Real Roughness $[\mu]$
Re	Reynolds number $[-]$
T	Temperature $[K]$
t_b	Burning time $[s]$
$t_{channel}$	Space between the cooling channels $[m]$
th	Thickness $[m]$
u	Velocity $[m/s]$
V	Volume $[m^3]$
VR	Volume ratio $[-]$
z	Altitude $[m]$
η_{junc}	Junction Efficiency
ρ_0	Dimensionless Radius $[-]$
σ	Stress $[MPa]$

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Chapter 1

Introduction

1.1 Problem Overview and Data

This report examines the performance of a liquid propulsion system employing a blowdown architecture, with the combustion chamber pressure ranging between 50 to 20 bar.

The system is fueled by a combination of liquid oxygen (LOX) stored at 90 K and RP-1 fuel at 300 K. The initial required thrust output is precisely specified at 1000 N, representing nominal conditions in vacuum,

neglecting thrust losses for the design process, considering them a posteriori in the results analysis.

Designed with a cylindrical configuration, the space allocated to the propulsion system features a diameter of 1 meter and a length of 2 meters. Despite this, 20% of the total volume of the case must not be occupied by the combustion chamber or the tanks to accommodate essential components such as pipings, wirings, and auxiliary systems. This analysis aims to show the correlation between a blowdown architecture and the variation of the operational characteristics and thermal dynamics, crucial for optimizing the performances.

$P_{C_{initial}}$	50 bar
L	2 m
ϕ	1 m
T	1000 N

Table 1.1: Given data

1.2 Assumptions

It is necessary to impose certain assumptions to simplify the calculation process without limiting the validity for a preliminary design:

- The behavior of the gases in both the tanks and combustion chamber is assumed to be that of an ideal gas;
- A frozen equilibrium condition is considered in the combustion chamber to prevent further reactions in the nozzle. The flow in the nozzle is assumed to be isentropic;
- The temperature in the combustion chamber is assumed to be constant, as well as the parameter c^* ;
- The combustion chamber is assumed to have a cylindrical shape;
- The combustion chamber and the nozzle are made of Titanium Grade V [4].
- The tanks are assumed to be perfectly cylindrical for volume calculation purposes and made of Carbon Fiber [5].

M_{mol} [g/mol]	ρ_{RP-1} [kg/m ³]	ρ_{LOX} [kg/m ³]	O/F_{ratio} [–]	γ_{cg} [–]	L^* [m]	ε [–]	T_c [K]	ε_c [–]
21.9	810	1140	2.24	1.24	1.08 [6]	70	3571	11

Table 1.2: Assumed data

The first 8 values in this table are found in [7], while the last one [8]

Chapter 2

Literature Review

2.1 Nozzle Losses

Compared to an ideal nozzle, real nozzles have losses and some of the internal energy is unavailable for conversion into kinetic energy of the exhaust. These losses may be taken into account in two ways:

- Using empirical correction factors based on experimental data;
- Developing more accurate algorithms.

Generally, nozzle analysts prefer to account for the many types of nozzle thrust losses by applying correction factors to the performance characteristics of an ideal nozzle.

The loss mechanisms fall into three categories: 1) geometric or divergence losses, 2) viscous drag losses and 3) chemical kinetic losses [7].

Divergence losses represent one of the main loss sources for real nozzles. The exit flow includes a non-axial component involving a loss of performance, because the radial velocity component absorbs energy to expand but does not produce thrust. This aspect can be taken into account with an analytical correction factor: the ratio between the integrated axial components of a real nozzle and the ones of an ideal nozzle. These divergence losses are typically small for most nozzles (1%), but may present a sizeable fraction of the total nozzle performance loss and may be quite significant in nozzles with large divergence angles. Divergence losses also strongly depend on nozzle geometry. The simplest nozzle configuration possible is the conical nozzle, where the flow diverges so that it is no more aligned with the axis and the divergence angle is maintained constant. In order to take this loss into account, it is necessary to introduce a dimensionless parameter, called the *divergence loss coefficient* λ , which can be computed through 2.1:

$$\lambda = \frac{1 + \cos \alpha}{2} \quad (2.1)$$

where α is the divergence half angle.

Although the conical nozzle represents the simplest configuration usable, most space propulsion systems are characterized by a bell configuration, which is constructed to have the velocity vector at the exit parallel to the nozzle axis. This configuration allows to minimize the loss due to the divergence of the flow. Since this configuration is complex to derive in a preliminary design analysis, parabolic nozzles, that represent an approximation of this type of bell nozzles, can be taken into consideration. One particularly common nozzle design is the RAO nozzle, that is a method to design the nozzle contour to yield optimal thrust. In order to design this type of nozzle, the angles at the beginning and at the end of the parabolic contour have to be known: they depend on the expansion ratio of the system and can be found using Figure A.2. They are calculated between the axis of the nozzle and the lines tangent to the parabola at initial and final points of the divergent part. The RAO approach takes as reference a conical nozzle with half angle of 15° and refers the length of the new nozzle to it. There exist two different RAO methods: the maximum performance approach and the minimum length approach. Differences between these methods regard the length of the nozzles and the thrust they can provide. For this type of nozzle, the divergence loss factor λ can be computed taking into account also the exit angle of the parabola:

$$\lambda = \frac{1}{2} \left[1 + \cos \left(\frac{\alpha + \theta_e}{2} \right) \right] \quad (2.2)$$

Viscous drag losses are due to the fact that real nozzles always develop thin viscous boundary layers adjacent to their inner walls, where gas velocities are much lower than in the free-stream region. Immediately next to the wall the flow velocity is zero; as you move further from the wall the speed increases until the free-stream velocity is reached. Boundary layer loss is a result not only of friction effects, but also of the heat transfer with the combustor wall, as can be shown in Appendix A.1. The boundary layer effect causes an apparent shrinkage of the nozzle cross section, due to the presence of the so called *displacement thickness*. As a way to take these

losses into consideration, a discharge coefficient can be introduced, depending on a modified Reynolds number of the flow (computed as $Re' = Re\sqrt{r_t/r_c}$, where r_t is the throat radius and r_c the throat curvature). This correction factor is defined as:

$$C_n = \frac{\dot{m}_{real}}{\dot{m}_{ideal}} \quad (2.3)$$

As described in [9], these losses represent an important problem for small modified Reynolds flows, while increasing this value the flow becomes more and more inviscid, so friction effects become negligible.

Chemical kinetic losses regard the combustion process, which can never be totally steady. Gas properties will oscillate with time and will not be necessarily uniform across the flow channel. Moreover, gas composition may change inside the nozzle when chemical reactions are occurring in the flowing gas. For this reason, the assumption of chemically-frozen-composition gas flow may not be fully valid for all chemical systems. These energy losses due to non-uniform unsteady burning are difficult to assess theoretically. The only way to decrease this loss is to reduce the nozzle exit temperature, but generally this causes the rise of other important losses.

2.2 Additive Manufacturing

Metal additive manufacturing has become more and more important through the years. The aim of this review is to summarize the state of art about uncertainty prediction and estimation for the main metal-AM techniques.

2.2.1 Overview On AM Techniques And Issues

The current additive manufacturing market comprehends many different techniques, but standing to the studies retrieved from the literature the most used and studied technique appears to be the *Powder Bed Fusion (PBF)*, specifically *Laser Beam (LB)* and *Electron Beam Melting (EB or EBM)*. PBF system is capable to produce high resolution components with complex internal geometry. It exploits an enclosed chamber at very low pressure, a heat source (laser or electron beam), a powder bed, a powder supplier and a scanner. The heat source only melts the powder required to build the structure while the rest acts as a support to the structures[10]. PBF handles metal powder with a particle size ranging between $20\ \mu m$ to $60\ \mu m$ producing consequently a higher or smaller resolution of the part [11]. Additive manufacturing is affected by its nature from [3] [11]:

- **Item orientation with respect to the bed:** together with its geometry it determines the imperfections in the shape, in particular with sharp edges and small round shapes, whenever those fall in the same dimensional range of the powder grains;
- **Item geometry and size:** when building large objects with wide continuous surface, they can act as heat sinks, releasing then the heat and interfering with the process in the next layers;
- **Powder size:** coarser powder can cause issues with smaller geometries;
- **Beam spot size:** the smaller the spot, the higher the resolution;
- **Scanning pattern:** there is wide agreement in the literature about the necessity to use a specific path that avoids the heating of the same spot with the same timing in subsequent layers; this to avoid overheating in those areas;
- **Layer thickness;**
- **Cooling time.**

2.2.2 The Uncertainty Quantification Problem

Additive manufacturing is a relatively new but extremely innovative technology. It demanded to rethink the design procedure of the pieces, but this also requires new tolerance assessment and estimation procedures. On one side the layer-wise manufacturing introduces a new parameter, the layer thickness, that should be distinguished from the deformation in the shape, being its nature different and repeatable [3]. The literature still lacks a specific method for the accuracy evaluation of the techniques, but most of the existing methods rely on *benchmark pieces*. In the following, two approaches are analysed.

Assembly Benchmark

In the paper *Geometric tolerance and manufacturing assemblability estimation of metal additive manufacturing (AM) processes* [11], an attempt of defining proper tolerances for assemblies manufacturing is attempted. To do so, two complementary items are developed (in Figure B.1). Both are characterized by a lattice structure and two pins and two holes, respectively. They are thought to be joined in a single assembly. The items are then manufactured in Inconel 718 with LPBF technology [11].

The study tries to quantify the geometric behaviour of the individual components of the assembly and then define the tolerances for the assembly, defining a *predictive skin model*. This review aims to analyse the results in terms of uncertainties and tolerances, the model itself is not reviewed.

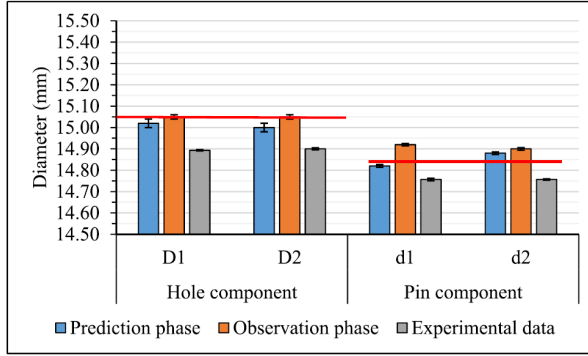


Figure 2.1: Comparison of diameter values

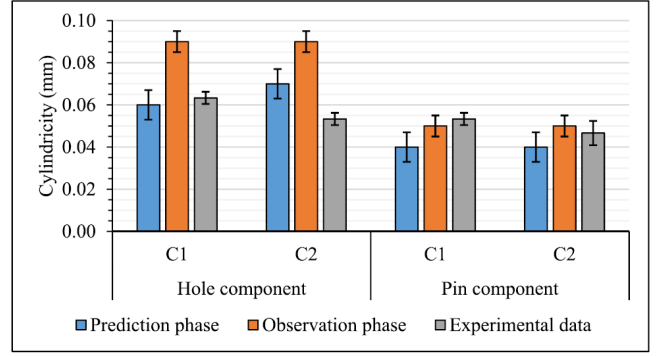


Figure 2.2: Comparison of *cylindricity*^[B.1.1] tol.

The nominal diameter of the pin is $\phi_{pin} = 15.055 \text{ mm}$ while for the hole it is $\phi_{hole} = 14.840 \text{ mm}$ in order to allow the coupling of the assembly. The diameter of the manufactured piece tends to shrink for both the pin and the hole. Overall an average surface roughness of $Ra = 0.0165 \text{ mm}$ and an average tolerance of $tol = 0.100 \text{ mm}$ with a minimum tolerance of $tol = 0.04 \text{ mm}$ [11].

Features Benchmark

Gruber *et.al* [1] developed a newly designed test shape to evaluate the capabilities of any metal-AM techniques of printing particular features as holes, cooling channels of different size, overhangs, hollow cylinder and more (Figure B.2). It is designed according to the following requirements [1]:

- different geometrical elements;
- different feature size;
- surfaces facing up and down;
- sufficient spacing between the features to allow 3D scanning;
- internal features requiring powder removal after the manufacturing;
- feasible and scalable for all AM processes.

After the manufacturing of the bench pieces with different techniques (Laser Metal Deposition (LMD), Laser Powder Bed Fusion (LPBF) and Electron Beam Melting (EBM)), it is performed an analysis of the results in terms of uncertainties through a 3D scan of the piece and the subsequent comparison with the CAD model. Figure 2.3 presents the results of the scan through a color scale where:

- **green:** small deviation from the CAD
- **blue:** negative deviation from the CAD
- **orange/red:** positive deviation from the CAD
- **purple:** large deviation from the CAD

Overall LPBF presents smaller deviations in the order of $150 \mu\text{m}$ with peaks to 0.5 mm for smaller features and overhangs. EBM and LMD shows deviation between $250 \mu\text{m}$ and 1.5 mm . EBM poorly performs on the small and thin feature as hollow thin cylinders and thin walls, where it presents the highest deviations (*red areas*). LMD is less effective than LPBF but still performs better compared to EBM [1].

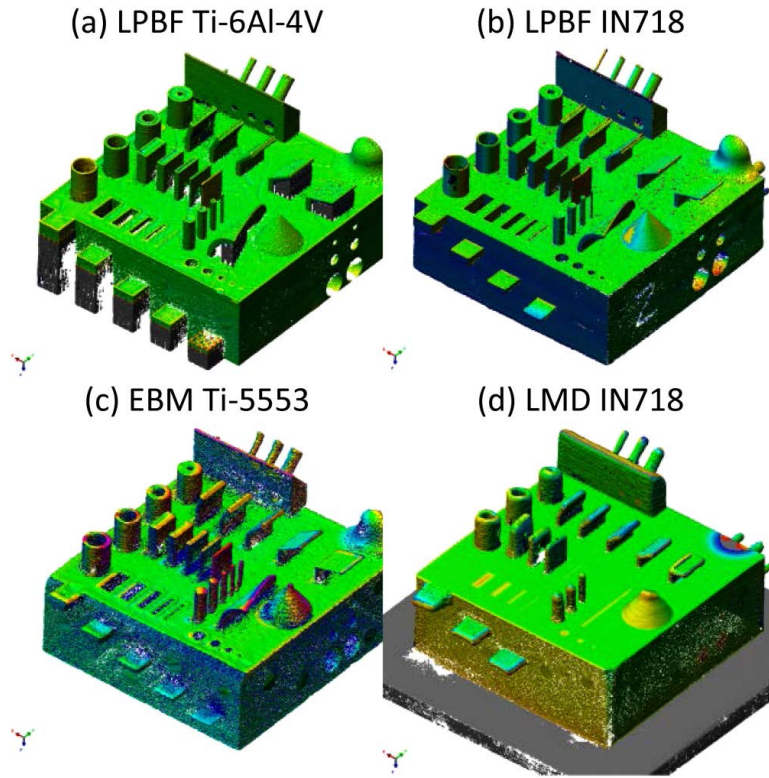


Figure 2.3: 3D scan of the benchmark piece, presenting the deviations from the CAD design [1]

Estimation of ISO Tolerances For Metal AM

The only study that tries to adopt the currently available regulations (namely the ISO 286-1 [12]) is the *Systematical Determination of Tolerances for Additive Manufacturing by Measuring Linear Dimensions* [3]. Of the techniques treated in this review only LPBF (*LS in the paper*) performed here with stainless steel 316L is analysed in the paper. In Figure B.3 a summary of the results of AM compared to *ordinary* techniques is reported. Overall the LS technique presents an IT class ranging between 13 to 15.

Aerospace Application: Injection Plate Design - Experimental Results

Moreover, two tests of an injection plate designed for AM are here considered. The first study from *Soller et al.* [2] highlights the major impact of the production of the raw material (powder) and its properties on the final result, and quantifies the impact in terms of uncertainty in the performance. They produced the plate with Inconel 718, which from the literature analysis appears to be one of the prime materials used in these applications. The study specifically tried to assess the performance of the injector without applying any post-manufacturing finishing. It could not present any data in terms of specific tolerances, but it could assess a certain reproducibility of the performance. The scattering in the discharge coefficient for fuel and oxidizer line is 3% and 4%, respectively.

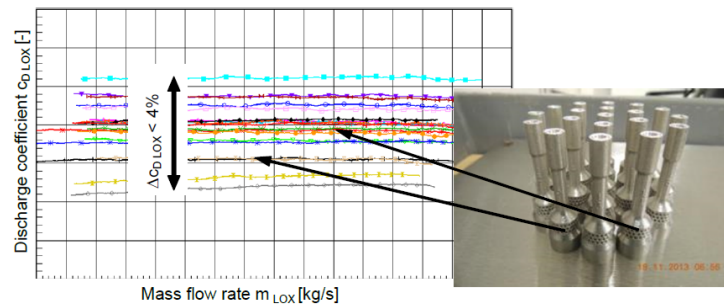


Figure 2.4: Scattering of the C_d values for the LOX orifices [2]

The second study from *Gloria et al.* [13] assessed the performances of their injection plate, in terms of shift in the $C_d A$ with respect to the design point. They performed few iterations achieving a final design of the injection plate made of Inconel 718 and characterized by orifices with a diameter of $\phi_{fuel} = 0.71 \text{ mm}$ and $\phi_{oxidizer} = 1.29 \text{ mm}$ and discharge coefficients $C_{d,fuel} A = 2.31 \text{ mm}^2$ and $C_{d,ox} A = 31.2 \text{ mm}^2$. The paper presents a shift of the $(C_d A)_{fuel} = -14\%$ [B.1.3] and $(C_d A)_{oxidizer} = +8\%$ [B.1.2]. The cause of the shift regarding the fuel line is to be attributed to the internal roughness of the orifices, while for the oxidizer side further investigation should be performed, mainly on how the N_2O_4 (here used as oxidizer) achieves the equilibrium inside the orifices [13].

2.2.3 Final Comment

The two studies cited above ([11] and [1]) are here analysed together with others ([3], [10], [13], [2], [14]) to present a comprehensive overview of the *state of the art* of tolerances and uncertainties estimation in metal additive manufacturing.

The uncertainties in the metal additive manufacturing appear to be highly inconstant due to the strong dependence of the print outcome on many geometrical and process factors, the material used and the particle size. At a first glance, the metal-AM may appear unsuitable for precision manufacturing due to the IT classes ranging between 13 to 15 [11]. By the way, the study computing those parameters was performed in 2015 and only on one metal-AM technology, meanwhile, the additive manufacturing has advanced. Standing to most recent studies, additive manufacturing appears to be much more precise, and many different prediction models are now available. According to *Gruber* [1] the tolerances for metal additive manufacturing techniques range respectively:

- **LPBF** (*Laser Powder Bed Fusion*): mainly *f* (*fine*) class for small features between $0.5 - 6 \text{ mm}$ (B.1.4);
- **EBM** (*Electric Beam Melting*): mainly *m* (*medium*) and *c* (*coarse*) class;
- **LMD** (*Laser Metal Deposition*): large range of possible deviations thus all class of tolerance are possible.

It is also understood that the highest accuracy is achieved with Ti-6Al-4 V instead of the more common Inconel 718. Apparently the difference between LPBF and EBM/LMD has to be searched in the optimized parameter set and the smaller powder particle, laser beam spot diameter and layer thickness [1]. Moreover LPBF is better for complex inner structures since EBM tends to leave powder remains inside the structures. According to the literature review, the best technique, in order to achieve the finest results without any mechanical reworking, is LPBF together with Ti-6Al-4 V or Inconel 718 (the latter already widely used).

2.3 Blowdown Architecture

In order to supply a liquid rocket motor with propellant, two different methods can be used: the pressure-fed and the turbopump-based. The first approach involves pressurizing a liquid container with an inert gas. This gas occupies a space over the liquid level of the vessel, known as ullage. The ullage tends to expand and push the incompressible liquid propellant out of the exit, inside the feeding lines of the propulsion system. There exist two possibilities to obtain a pressure-fed method: the pressure regulated and the blowdown system [15]. The main feature of a blowdown system is that the pressurized tank contains both pressurizing gas and liquid propellant since the beginning, without pressure regulation. This design choice avoids the complexity of a separate pressurant tank and, hence, improves the reliability. Pressurisation is guaranteed by the only presence of the gas in the tanks: as it expands it is losing pressure, thus lowering the pressure in the tanks. Blowdown systems are utilized in both monopropellant and bipropellant propulsion units, and over the years, specific technologies for tank repressurization have also been developed.

This particular configuration represents an attractive alternative to the more complex and expensive pressure regulated systems. For example, Magellan spacecraft's primary propulsion unit used this type of blowdown system, provided of an additional helium tank for the repressurization of the single hydrazine tank [16]. Moreover, throughout the years also bipropellant blowdown systems with repressurization have been developed, as stated in [17]. High-pressure spherical tanks are placed upstream of the propellant tanks to provide repressurization

by firing pyrovalves when the tank pressure decays to a prescribed value. Figure 2.5 shows the development of these possibilities in common and separate repressurization configurations. Repressurization of the propellant tanks allow for a greater propellant loading efficiency while maintaining an inlet pressure consistent with thruster capabilities.

The figure on the left shows a configuration which involves simultaneous repressurization of the fuel and oxidizer tanks when the pressure in one of the tanks reaches a specified minimum value. Either increasing the pressure blowdown range or adding a second repressurization would allow an increased propellant load. On the right it's shown a configuration that involves the separate repressurization of each tank when a minimum pressure is reached in either one. This concept also has an 80% propellant loading efficiency and provides complete isolation of fuel and oxidizer while eliminating the need for check valves [18]. The blowdown system is also self-regulating since a more rapid pressure decay will bring the system back toward its design point mixture ratio. The feasibility of a bipropellant blowdown system depends on the ability of the engine to run stably and with proper thermal control over a range of feed pressures and mixture ratios.

This system is mainly utilized in orbit adjust thrusters. The control of oxidizer and fuel flows is fundamental and is provided generally by orifices and venturi systems. The orifice control maintains the same mixture ratio into the thruster for equal fuel and oxidizer inlet pressures. On the other hand, a venturi induces a slight shift in the mixture ratio but offers the advantage of reducing the total range of flow rates into the thruster. Incorporating a cavitating venturi further enhances the stability of the feed system, since it is very effective in decoupling downstream pressure oscillations from the upstream flow [19].

Blowdown configurations are generally characterized by a certain hazard in failure or rupture of pressurized vessels. Therefore, it becomes crucial to control blowdown process to prevent such accidents. The critical parameters in blowdown design are the selection of construction material, the sizing of relief valves, temperature and pressure [20].

Different types of pressurant gases can be adopted in a blowdown system, but the most common are nitrogen and helium. In Figure C.1 are shown the different behaviours between these two pressurants. The lower initial pressure relative to the loading pressure is a result of gas solubility. The high solubility of nitrogen is especially evident, and the dispersion between fuel and oxidizer pressures illustrates the disadvantages of using nitrogen in this type of system. Helium also appears to be a better pressurant because it exhibits less variability and less sensitivity to changes in the operating environment. For these reasons, helium as pressurant is almost mandatory.

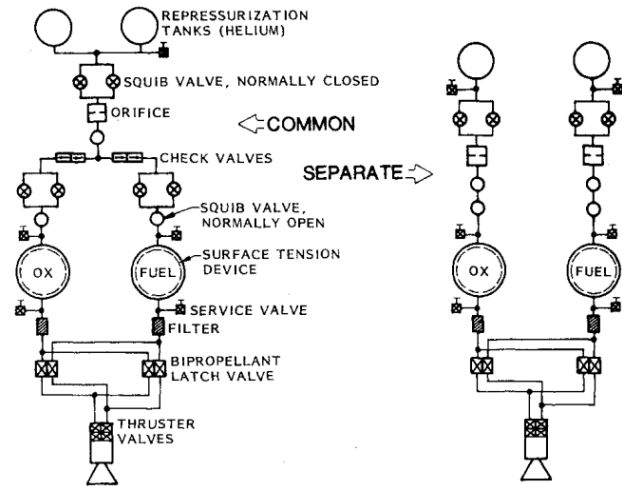


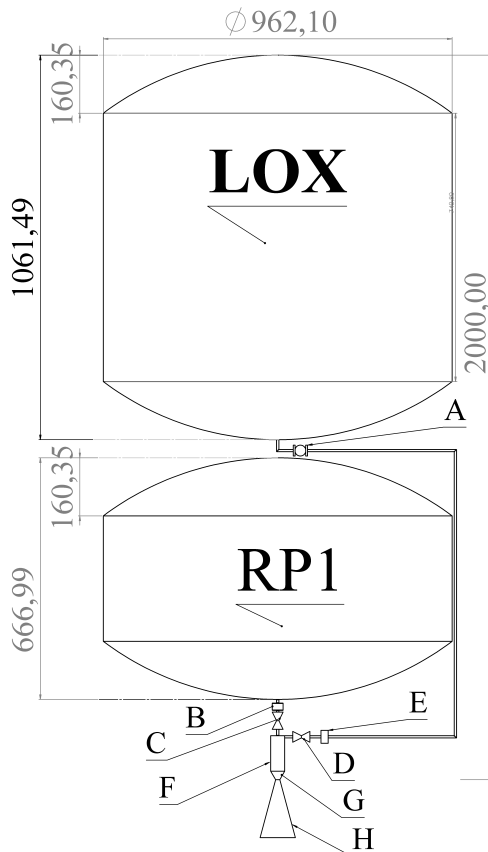
Figure 2.5: Common and separate repressurization configurations

Chapter 3

Architecture and Materials

3.1 Architecture

3.1.1 Architecture Overview



- A : Positive Shut-Off Valve
- B : Filter
- C : Latch Valve
- D : Ball Valve for Cryogenic Applications
- E : Filter
- F : Combustion Chamber
- G : Convergent Duct
- H : Divergent Nozzle

Cylindrical vessels in tandem configuration are selected to be compliant with the size constraint. They are made of carbon fiber which lowers the overall weight and it is compatible with both LOX and RP1. Piping and valves in Titanium Grade II which is compatible with the carbon fiber tanks and the low temperature of cryogenic oxidizer. The thrust chamber is made of Titanium Grade V (Ti-6Al-4V) for enhanced precision during additive manufacturing and better compatibility with the pipes.

Figure 3.1: Architecture Overview

3.1.2 Sizing of The System

The optimization of the architecture is performed aiming to an optimal 80 % filling of the case, leaving 20 % of empty spaces for pipes and subsystems. In order to pursue a simple architecture, *exotic* tank shapes are discarded, thus the volume around the combustion chamber cannot be used for propellant allocation. That volume is quantified as 6 % of the total, so the remaining 14 % has to be distributed around and in between the vessels. A longitudinal distance of 5 *cm* between fuel and oxidizer tanks and of 10 *cm* is considered sufficient according to the expertise to locate pipes, valves and filters. Accounting for this extra empty volume already assigned, the remaining percentage is distributed uniformly around the tanks, obtaining the final length and diameter of each.

3.1.3 Igniter

A pyrogen igniter is evaluated, since it is the typical hardware used in liquid propulsion. It provides several advantages as a relatively low price per ignition and a compact configuration, despite the main disadvantage concerning the possibility of only one single ignition per device. Another possible technology is the spark torch igniter that is characterised by re-ignitability and clean combustion products, but on the other side, it needs high voltage electronics parts and feed systems for oxidizer and fuel. In addition to the previous aspects, the spark

torch igniter is ranked as TRL-6 technology, while pyrogen igniter is TRL-9, so it is already operative [21]. For all these reasons, the selected technology results to be the pyrogen. It is possible to calculate the mass of the igniter knowing the volume of the combustion chamber and exploiting an empirical formula obtained from experimental data (F.4): $m_{igniter} = 0.296kg$. It needs to be said that, after having computed the dimensions of the combustion chamber, placing a pyrogen igniter inside the small chamber could create some issues on the required size of the igniter.

3.1.4 Pipes, Valves and Filters

Pipes

Line	Length [m]	Diameter [m]	Section area [m ²]	Roughness [μm]
RP-1	0.100	0.0064	$3.1672 \cdot 10^{-5}$	2
LOX	1.7291	0.0063	$3.1662 \cdot 10^{-5}$	2

Table 3.1: Pipes sizing

Pipes are made of Titanium Grade 2, which is capable of withstanding the low temperature reached by LOX preserving its mechanical properties; moreover, it significantly limits the galvanic corrosion problems when in contact with the carbon fiber tanks [22]. According to the consideration of *Levesey et al.* [23] also Titanium Grade 6 (Ti-5Al-2.5Sn ELI) may be evaluated thanks to its improved mechanical behavior at cryogenic temperature.

Filters

Fuel and oxidizer lines both include **filters** immediately before the valves to stop any debris coming from the upstream section of the pipe due to residues of the manufacturing process.

Valves

The **valve** selection is done with different criteria for the fuel and oxygen lines.

CAD Label	Line	Type	Model	Proof Pressure [bar]	Burst Pressure [bar]	Operating T [K]	Average P Drop [bar]	Material	Activation
C	RP-1	Latch Valve	Vacco V1E10763-01	345	517	263.15 ÷ 333.15	0.69	Titanium	Electrical
A	LOX	Positive Isolation Valve	Vacco 1104900-01	344	517	-	-	Titanium	Electrical
D	LOX	Ball Valve	in-house	60	120	65 - 100	~ 0.1	Titanium	Pneumatic

Due to the peculiarity of the cryogenic oxidizer feeded at 50 bar, it is not possible to select a commercial valve, thus an *in-house* manufacturing is advised. According to a market review it is suggested to use Titanium Grade 2 which is capable to withstand pressure and temperature of the LOX. Moreover, a ball mechanism appears to be the most common on currently available valves for cryogenic fluid. An alternative may be the Marotto PV20T specifically meant for cryogenic and high pressure applications, but material compatibility should be assessed, since it is made of *Cu/Ni* and *Tin Alloy*. Before the ball valve there is a positive insulation valve (normally open) which acts similarly to pyrotechnic valves but without any pyrotechnic action; it is indeed electric, enhancing the safety of the system. It is meant to provide a quick shut off of the engine in a fuel rich environment so that heat spikes are prevented [7].

3.1.5 Diaphragm

In order to avoid dissolution of helium in the liquid propellant and bubbles formation, it is useful to insert inside the tanks a diaphragm, which separates the ullage gas from the propellants. For the selection of the material, it is necessary to consider the low temperature of the oxidizer that does not have to limit the mechanical properties of the diaphragm and the compatibilities with liquid oxygen, RP-1 and helium. Rubbers and elastomers are typically used for these applications and could be compatible with helium and oxidizer, but their validity for RP-1 is not guaranteed [24]. Teflon is a very non-reactive material, which maintains high strength, toughness and self-lubrication at low temperatures [25]; it results to be compatible with both propellants and helium [26] [27] [28]. The ultimate tensile strength with respect to the operating temperature is reported in Figure F.2.

3.1.6 Tank Architecture

Even though the spherical geometry offers the smallest surface-to-volume ratio, a cylindrical shape for the body of the tanks is selected due to the constraints on diameter and length of the case [29]. The tandem disposition allows to put the oxidizer tank far from the combustion chamber, in order to avoid that liquid oxygen gets heated. To ease volume and mass computations, the heads of the cylinders are considered flat, while for the strength assessment the proper elliptical head are considered. The elliptical head choice is guided by the necessity of sustaining considerable pressures that would concentrate in the sharp edges of a flat head. In order to reduce the volume loss when moving from the flat approximation to the rounded one, keeping the advantages in terms of strength, the head height is set to $\frac{1}{6}\phi_{tank}$.

	L_{total} [m]	ϕ [m]	h_{head} [m]	V_{cyl} [m ³]	th_{cyl} [mm]	th_{ellip} [mm]	m [kg]	Material
Fuel	0.667	0.962	0.160	0.485	1.1	0.71	1.238	Carbon Fiber
Oxidizer	1.061	0.962	0.160	0.772	1.1	0.71	1.971	Carbon Fiber

Carbon fiber is selected as the material for the tanks, despite not being suitable for additive manufacturing; it produces a major weight and thickness reduction, the latter is critical to comply with dimensional constraint. In order to be more conservative the lower end of the ultimate allowable stress for carbon fiber is selected, reading $\sigma_{ultimate} = 2.5 \text{ GPa}$. The thickness assessment is performed according to *ASME UG-16* [F.1]. It is assumed full efficiency ($\eta_{junc} = 1$) of the head junction (being a continuous fabric) and a $P_{burst} = 120 \text{ bar}$, twice the expected operation pressure. For a safer approach it is considered a vessel with a constant thickness computed with the equation for the cylindrical case [F.3]. For a weight saving approach it is suggested to diminish the thickness of the elliptical heads adopting th_{ellip} .

3.2 Material Compatibility: A Summary

Cryogenic propellants can cause tank design problems due to high thermal gradients, thus they need, for construction, materials capable of remaining ductile at very low temperatures [29]. Since the oxidizer is liquid oxygen, there can be also issues concerning the chemical compatibility with materials, which shall not oxidize during the mission. The high corrosivity of the oxidizer represents a strong restriction for the material selection: stainless steels and titanium are usually the most common choice for applications similar to the analyzed one, but also carbon fiber is recently being studied as tank material for cryogenic propellants. It presents the same mechanical properties of the stainless steel in traction, with the main advantage, with respect to stainless steel, of a lower density. Carbon-fiber tanks that contain liquid oxygen have been used in some recent space missions [30] [5], moreover the carbon fiber results to be compatible also with RP1 [31]. In order to maintain the temperature of liquid oxygen below its boiling point, it is preferable to add an insulant coating to the oxydizer tank. As already mentioned, Titanium II is selected for piping and valves, since it is the material that best suits the low temperature of LOX and also limit the galvanic corrosion when in contact with the carbon fiber tanks. Valves are made in titanium as well to avoid any possible corrosion when coupled with the pipes. Finally, after a first iteration considering Inconel 718 for the combustion chamber, Titanium Grade V (Ti-6Al-4V) is chosen, to provide better compatibility and optimal performance during the additive manufacturing process [1] [4].

Chapter 4

Modeling Approach

4.1 Nominal Design

4.1.1 Performance Parameters

In the initial phase of the analysis, isentropic thermodynamic relations are employed to calculate the fundamental thermodynamic characteristics of the design. These fundamental parameters serve as the basis for determining the overall configuration of the engine and are derived by taking into account the thrust requirements and the assumed expansion ratio. In this design, the requirement of achieving a thrust of 1000 N is thrust is considered to be true before applying any corrections for losses 2.1. Utilizing the established isentropic relationships 4.15, D.5, D.6, D.7, D.8 the fundamental parameters are obtained to initiate the design process. These values can be found in 5.1.1

4.1.2 Combustion chamber design

In this application, the initial nominal chamber pressure is utilized to design the combustion chamber, although it varies between 50 and 20 bar due to the blowdown architecture. Moreover, for space applications where ambient pressure is negligible, maintaining a high pressure within the combustion chamber is unnecessary. The choice of a cylindrical design and Titanium grade V as material for the combustion chamber is driven by the compatibility with additive manufacturing processes, which allows an easier fabrication and improved the performances. Additionally, the incorporation of RP-1 cooling jacket channels within the chamber enhances heat dissipation, further justifying this selection. The characteristic length L^* plays a significant role in its design, defined as:

$$L^* = \frac{V_{cc}}{A_t} \quad (4.1)$$

The selection of L^* is guided by propellant considerations, with a value of 1.08 m employed in this instance [7]. By computing the throat area D.9 establishing the characteristic length, the combustion chamber volume can be derived using the inverse formula. Additionally, a contraction ratio $\varepsilon_c = 11$ is enforced to determine the chamber's area and diameter. To ensure operational efficiency and stability, it is fundamental to maintain and verify that the Mach number within the combustion chamber is below 0.4, so as to minimize pressure losses at the entrance of the chamber and guarantee adequate mixing of the fluid and uniform distribution. The correlation between volume and area permits the computation of the actual combustion chamber length 5.2.

4.1.3 Nozzle Design

As explained in Section 2.1, the nozzle geometry design is fundamental in order to compute the thrust losses. Different methods have been taken into account: the simple conical geometry and the RAO approach. Divergence loss coefficients are computed using equations 2.1 and 2.2, for conical and RAO nozzle respectively. Rao configuration can be divided into two different methods: the minimum length approach (MLA) and the maximum performance approach (MPA). In the first method, the length of the divergent part of the nozzle is minimized and fixed equal to the 60% of the length of a 15° conical nozzle. The initial and final wall angles are respectively $\theta_i = 38^\circ$ and $\theta_e = 13^\circ$, extrapolated from the RAO curves, with the considered expansion ratio, reported in Figure A.2. The second method regards a nozzle with equal length to the one of the 15° conical nozzle. The wall angles are $\theta_i = 29^\circ$ and $\theta_e = 4.5^\circ$. The aim of this analysis is to minimize these losses, thereby achieving a thrust profile that is closer to the nominal one as much as possible. For this reason, the chosen configuration is the maximum performance approach, which is able to provide the minimum percentage of thrust losses. The thrust obtained considering the divergence losses can be computed through the equation:

$$T_{2D} = \lambda \dot{m}_p u_e + (P_e - P_a) \cdot A_e \quad (4.2)$$

where $\dot{m}_p$ is the propellant mass flow rate, u_e is the flow exit velocity, P_e is the exit pressure, P_a is the ambient pressure (in this case it is null, since the analysis is realized in vacuum) and A_e is the nozzle exit area. The loss coefficient enters only in the dynamic term of the thrust, because the static contribution only depends on the exit pressure and area.

4.1.4 Pressure Cascade

For the pressure cascade four different types of pressure drops are considered:

- **Dynamic:** corresponds to the pressure drop due to the startup of the fluid exiting the tank. For the computation, the *Bernoulli* equation is used.
- **Feeding lines - Distributed:** starting from the data reported in the references [32][33], which include the correlation between pressure and viscosity for both LOX and RP1, it is possible to interpolate the trend of viscosity within the pressure range used by the tanks designed in section 4.1.5. Knowing the architecture, the material of the pipes, and their area, it is possible to calculate the velocities in the pipes using the continuity equation. Given these parameters, it is possible to calculate the Reynolds number associated and the real roughness as:

$$Re = \frac{\rho \cdot u_{pipe} \cdot \phi_{pipe}}{\mu} \quad (4.3) \quad Ra_{real} = \frac{Ra}{\phi_{pipe}} \quad (4.4)$$

These two values are essential to derive the friction factor f using the Moody diagram reported in Figure E.2. This procedure is carried out using 10 pressure tank values obtained from the discretization during the burning time. A subsequent discretization of the ten found Darcy factor values is necessary in order to calculate, at each instant of the burning time, the distributed pressure losses are equal to:

$$\Delta p_{distr} = \frac{1}{2} \cdot \rho \cdot \frac{L_{pipe}}{\phi_{pipe}} \cdot f \cdot u_{pipes}^2 \quad (4.5)$$

This procedure needs to be repeated both for the pipes containing fuel and those containing oxidizer, as they have different flow rates and their pipes have different diameters and lengths due to architectural choices.

- **Feeding lines - Concentrated:** for concentrated losses, the model that calculates the pressure drop at valve passage is used. For the calculation, the pressure variations with respect to $\dot{m}$ are found for each valve. Therefore, the different Δp are computed using an interpolation method.
- **Injection plate:** a modeling simplification is chosen, considering them as a percentage of P_c . For this project, it is chosen to consider them as 20% of P_c because the injection system is *like-on-like* [7].

The initial pressure in the tanks is set at 50 *bar* plus the pressure losses, while the final one (the guaranteed pressure when the tanks are empty) is assumed as 20 *bar* plus the initial pressure losses. This is an oversizing as the pressure drops will reduce with time as the velocity in the pipes is reduced.

4.1.5 Tanks and Pressurising Gas Design

Up to this part, the two tanks of the propulsion systems are sized as perfect cylinders, as explained in 3.1.2, since this simplification introduces only small approximation. A crucial constraint is the one specifying that the volume allocated to the propulsion system is 80% of the total volume occupied by the protective case, which is 2 m in length and 1 m in diameter. It is imposed that the wholeness of the assigned volume is occupied by the tanks and the combustion chamber, but the two tanks have not the same volume: this is because LOX and RP-1 have different densities and the *O/F* ratio is unbalanced towards the oxidizer.

The volumes taken by each of the tanks are computed in an ideal condition in which the oxidizer to fuel ratio is constant over time: to begin, the volumetric flows are computed from the mass flows, which are known; once these values are derived, the calculation of the volumetric ratio (volume of the oxidizer divided by volume of

the fuel) is immediate, and thanks to it a direct correlation that defines the distribution of the volumes between the two tanks with equation: $V_{tank,fuel} = \frac{V_{tot,tanks}}{1+VR}$

Once the volumes are set, it is needed to define the dimensions of the two tanks: length and diameter. In order to do this, it is needed to consider the constraint of a maximum total length, from the start of the tanks up to the throat of the combustion chamber, of 2 m; in this analysis it is also considered a 10cm separation between the fuel tank and the injection plate in order to leave space for the needed valves, as well as a 5cm distance between the two tanks, in order to allow space for the outgoing duct and, more importantly, a minimum separation between the tanks as they are kept at very different temperatures (this space allows some margin for the thermal protection around the oxidizer tank).

The total length of the combustion chamber up until the end of the convergent is known: this, after considering the aforementioned separations fixes the total length of the tanks. Knowing this, with simple geometrical formulas, it is immediate to derive the diameter of the tanks, which are considered equal; it is also important to check that there is enough space between the tanks and the case, particularly for the LOX tank, for the feeding lines, otherwise it would be necessary to consider an external piping system, and for the thermal coating. It is also immediate to derive the length of each tank.

Another crucial aspect is the sizing of the thickness of the walls, which depends on the material that is selected. After some attempts with different materials, it is decided that the best one in terms of thickness and weight is the carbon fiber.

The chosen material shows good properties both in terms of ultimate tensile strength (5300 MPa) and density (1770 $\frac{kg}{m^3}$). The thickness of the walls is set by fixing a burst pressure, typically assumed as double the maximum operating pressure in the tanks, which is computed by considering the initial pressure in the combustion chamber and adding the losses. As this value is around 60MPa for both tanks, the burst pressure is assumed to be 120 MPa; then, the thickness of the walls of the tanks is derived from 4.6.

$$th_{tank} = \frac{P_{burst}\phi_{tank}}{2\sigma} \quad (4.6)$$

This value will slightly reduce the available volume for what is stored inside the tank, as the thickness has to be considered as it is inside the total volume computed before, rather than outside, otherwise the constraint of occupying a maximum of 80% of the total volume of the case would be violated. Once the thickness, which is assumed to be the same for both tanks, is known, it is possible to obtain the masses of the tanks by multiplying its volume (a cylindrical crown) by the density of the carbon fiber.

Another crucial aspect for the design is how much mass of pressurising gas needs to be loaded, and the selection of the gas. The choice is between helium and nitrogen, and it is decided to opt for helium for the reasons explained in 2.3: this element has a specific heat ratio of $\gamma_{He} = 1.66$ and a molar mass of 4 $\frac{kg}{kmol}$. In order to obtain the mass of helium that is needed for the pressurisation, the method is the following.

The first step is to set the initial and final pressure in the tanks, considering the maximum and minimum values of combustion chamber pressure and adding the pressure losses, which can be identified as explained in 4.1.4. Once the initial and final pressures in the tanks are found, it is possible to compute the final temperature of the pressurising gas in the tanks; as a first approximation, the helium temperature is assumed to be the same as the one of the fuel for the RP-1 tank, and the same as the one of the oxidizer in the LOX tank. This is a strong approximation, as the temperatures will not be the same for the whole duration of the mission.

Once the initial temperatures are set (300 K for the fuel, which is storable, and 90 K for the cryogenic oxidizer), the final temperatures can be derived by considering an adiabatic expansion for the helium, specifically with eq.D.1 for the fuel tank and with eq.D.2 for the oxidizer tank.

Once the temperatures are computed, it is immediate to derive the masses of helium needed for each tank through the perfect gas law (the perfect gas approximation is used).

Furthermore, the initial volume of gas in the tank is computed through D.3 for the fuel tank and through D.4 for the oxidizer tank; these two equations represent adiabatic expansions.

From these values, it is possible to derive the effective volume of LOX and RP-1 that will be loaded, by subtracting the initial volume of helium from the total internal volume of the tank as 4.7 and 4.8.

$$V_{eff,RP-1} = V_{internal,RP1} - V_{He,in,RP1} \quad (4.7)$$

$$V_{eff,LOX} = V_{internal,LOX} - V_{He,in,LOX} \quad (4.8)$$

4.1.6 Injection Plate Design

Since the thrust and, as a consequence, the mass flow rates are quite limited (on top of this, it is expected that these values drop during the mission), it is decided that it may be convenient to select an injector with a small diameter, in order to ensure an easier flow atomization.

The injectors belong to the like-impinging family as the fluids are both injected in liquid phase; this type of injectors is also less sensible to instabilities with respect to the unlike family. In particular, a small tube with a conical entrance is selected, featuring a 0.5 mm diameter and a tabulated $0.7 C_d$ discharge coefficient. A brief consideration over the selection of the diameters of the injectors can be found in 5.1.5.

The first injectors to be sized are the ones for the fuel: the total area needed to discharge the nominal fuel mass flow rate is computed with 4.9.

$$A_{fuel} = \frac{\dot{m}_{fuel}}{c_d \sqrt{2\Delta P_{inj} \rho_{fuel}}} \quad (4.9)$$

Knowing this, after deriving the area of a single injector from its diameter, it is immediate to compute the number of injectors needed for the fuel through $N_{inj,fuel} = \frac{A_{fuel}}{A_{inj,fuel}}$.

A doublet configuration is assumed, so the number of the oxidizer injectors is the same as the fuel ones.

The discharge coefficient for the LOX injectors is assumed to be the same as the one for the fuel, as the mass flow of LOX is predominant, but the density of the oxidizer is higher than the one of the fuel: this terms balance each other in the formula that allows to compute the total area needed for oxidizer injection (4.9 with LOX mass flow rate and density). This allows to have a good balance between the critical issues rising due to both additive manufacturing and atomization. While the former matter requires the holes not to be too miniaturised, as the uncertainties would rise due to the use of this technology, the latter, as previously mentioned, imposes the holes not to be too large, as the injection plate is small, and a high area of the single injector could require a too small number of injectors, which could lead to a far from perfect atomization of the oxidizer flow.

Furthermore, it is needed to compute the exit velocity of the injectors in order to check if it is in the proper range; also, this values are needed to decide the injection angles of the two flows.

The exit velocities are computed through equation 4.10:

$$u_{exit,inj} = c_d \sqrt{\frac{2\Delta P_{inj}}{\rho}} \quad (4.10)$$

After this step, the design of the injection plate can be completed by deciding the injection angles. The criteria used to design this part is to obtain an *impinging angle* as close as possible to a value of 60° , as per [34]: it is stated that this value is a common one for impinging angles. This can be done by imposing an initial value for the injection angle of either the fuel or the oxidizer (in this case, the first one: $\delta_f = 30^\circ$), then computing the other one with 4.11, assuming a horizontal flow after the mixing and, finally, deriving the value of the impinging angle through simple trigonometrical considerations made considering Figure 4.1. The process is an iterative one, as the initial value of δ_f is updated to obtain an impinging angle closer to 60° .

$$\delta_{ox} = \arcsin \left(\frac{\dot{m}_f u_{exit,inj,f}}{\dot{m}_{ox} u_{exit,inj,ox}} \sin \delta_f \right) \quad (4.11)$$

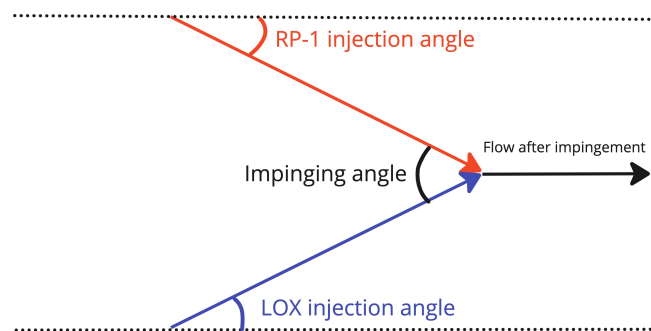


Figure 4.1: RP-1 and LOX injection angles are, respectively, δ_f and δ_{ox} ; the computation is based on obtaining an impinging angle as close as possible to 60°

4.1.7 Cooling System Design

In both the combustion chamber and the nozzle, extremely high temperatures are reached. To ensure the chamber wall material remains within its temperature limits and to prevent damage to both the wall and the chamber lining, a cooling system is essential. This system helps the mitigation of heat transfer and the integrity of the engine avoiding potential failures. The initial analysis focuses on the global perspective, and to streamline the process, the system is partitioned into three subsystems: Combustion Chamber, Nozzle Throat Section and Nozzle Exit Section. This methodology is consistently applied and repeated across all the sections to determine the total heat for each component. The general description of the passages is as follows: the radiative heat is assumed to be negligible, and the analysis concentrates on the convective heat exchange between the combusted gas and the chamber wall temperature. The utilization of the *NASACEA* software is necessary to derive the properties of the combusted gas, such as conductivity, viscosity, specific heat, and Mach number, for each section. With these known data, as listed in 5.9, Reynolds numbers and Prandtl numbers are computed through the equations 4.3, and the Nusselt number is determined using the Dittus-Boelter equation D.11. It is determined to use the Dittus-Boelter equation even though the ratio L_c/D is less than 10, since the other constraint regarding the Prandtl number is satisfied as it is greater than 0.6. By the inverse Nusselt number equation D.12, the gas-side heat transfer coefficient (h_{cg}), crucial for calculating the heat flux, is obtained. Additionally, by previously selecting Grade V titanium as the material for the chamber, the maximum operating temperature is obtained from its technical datasheet [4]. Subsequently, it is imperative to calculate T_{aw} , the adiabatic temperature correlated with T_c , as reported in appendix D.14, through the recovery factor r that represents the ratio of the frictional temperature increase to the increase caused by adiabatic compression D.13. The recovery factor. Ultimately, this enables to accurately quantify the adiabatic heat flux per unit area from the hot gases to the wall through D.15. It is chosen to use a regenerative cooling system to dissipate the heat, this technique is frequently used as method because it doesn't cause any performance losses, as the coolant is returned to the injector. Following this, a localized analysis is conducted to verify that the cooling capacity of the fuel is adequate to cool the wall temperature and avoid the boiling point. RP-1 is chosen exclusively for cooling purposes due to the cryogenic nature of LOX, which would become gaseous, despite its larger quantity. Additionally, the liquid oxygen tank is situated farther from the combustion chamber than the fuel tank, resulting in higher pressure losses. The adiabatic heat flux in the combustion chamber is calculated using D.16. Through relationship D.17, with all known values, it is possible to derive the temperature increase of the RP-1, ΔT_{RP1} . It is verified that the fuel reaches a temperature lower than that of coking T_{coking} [35] [36]. This implies that the choice of material for the combustion chamber and the quantity of fuel are suitable for cooling the combustion chamber. With the adiabatic heat flux established, it is possible to compute the minimum thickness of the chamber required to withstand the heat flux distributed by conduction through the walls of the combustion chamber, while imposing the temperature gradient (1600 K in the internal wall for thermal resistance of titanium and 450 K on the external wall to avoid vaporisation of RP-1) and the thermal conductivity of titanium, as follows in D.18. As a final step, the size and number of channels comprising the cooling jacket are determined. An initial geometry for these channels is hypothesized, and then, by calculating the flow rate in each channel, the Nusselt, Prandtl, and Reynolds numbers associated with the geometry are derived. It is necessary to iterate this process until finding a design that results in an acceptable deviation between the Reynolds number generated by the geometry and the one derived in the cooling jacket channels by imposing a fuel temperature lower than the coking temperature.

4.1.8 Cryogenic Tank Insulation

The design of an insulator is influenced by several aspects: an important one is the configuration, which can be internal or external with respect to the oxidizer tank. The internal configuration is characterized by a minimization of propellant losses during the tank filling, but on the other hand, it is characterized by a huge number of issues. Indeed the high pressure and temperature levels inside the tanks reduce the sealing capabilities of the coating and increase the probability of crack formation. Other problems, concerning the internal coating, deal with the storing period before flight and include difficulties in installation, locating, leak repairing and cleaning of the coating inside the tank. The external coating does not operate at the extreme conditions of the internal one and has the advantage of isolating the tank structure from the high temperature of aerodynamic heating during boost. Moreover, maintenance and sealing operations before flight can be done without any specific

consideration [29]. For all these reasons, the chosen configuration is the external one. One of the possible material for the insulator could be Cryogel Z, which is composed of a flexible aerogel blanket laminated to a vapor retarder. This configuration results to be perfect for cryogenic applications in space missions. It remains flexible, even at cryogenic temperatures, eliminating the need for complex and costly contraction joints, thereby resulting in simple and faster installation and it is designed for long term performance, while also withstanding incidental mechanical abuse, leading to continued protection through the mission time [37]. According to some sources, Cryogel z is still under evaluation for the rocketry applications, even if it has been widely used in several space applications for insulating purpose [37]. Another possibility is represented by polyurethane (PU) foam that guarantees good performances despite of some disadvantages as no re-usability, dust occurrence and hard managing insulation thickness [38]. Poliurethane foam has been used also for liquid oxygen tanks insulation of launch vehicles [39]. In order to compute the insulator thickness, it is important to know the heat flux impinging the spacecraft and the external temperature level of the rocket case. There are several papers and documents that treat the complexity of this problem and one of them is exploited to approximate the external temperature of the rocket case, considering the Sun the only power source. The insulator thickness is sized considering the maximum reached temperature and assuming that the spacecraft does not rotate [40]. The heat transfer from the external case to the insulator surface is supposed to be radiative and finally a thermal resistance approach is used to define the temperature differences between the insulator surface and the oxidizer tank inside.

$$T_{max} = T_0 \left[\frac{3}{4} + \frac{1}{8} \frac{\rho_0}{(\rho_0/2\pi)^2 + 1} \left(\frac{1}{e^{\rho_0/4} - e^{-\rho_0/4}} + \frac{\rho_0}{2\pi} \right) \right] \quad \rho_0 = \frac{r_s}{R} \quad (4.12)$$

where T_0 is the reference temperature.

The thickness of the rocket case is assumed negligible, thus the temperature of the inner case surface is equal to the external one; the temperature outside the insulator can be computed exploiting the Stefan-Boltzmann law. For the spatial distribution of the temperatures from the insulator to the inner part of the tank, the thermal resistance equation in cylindrical geometry are used:

$$\frac{\Delta T}{Q} = \frac{\ln(R_{ext}/R_{int})}{2 \cdot \pi \cdot k \cdot l} \quad (4.13)$$

The thermal conductivity of the two analyzed insulators have been interpolated from the curves F.3 and F.4. Finally the temperatures are plotted with respect to the radial position, as reported in Figure 4.2.

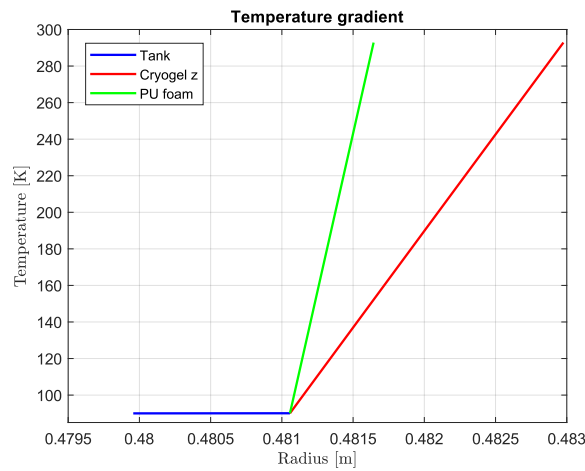


Figure 4.2: Thermal gradient with different insulators

4.2 Time Profile

Once the initial conditions are established and the tank design is determined, the time profile of the parameters can be modeled. To predict the evolution of these parameters over time, the necessary initial conditions include

the pressure inside the tanks at the initial and final instants. At the final instant it is assumed to have the minimum pressure available in the chamber (20 bar) plus the total pressure drop calculated at the initial condition, as specified in section 4.1.4. Other required parameters consider the tank inner volumes, as outlined in section 4.1.5.

The adiabatic relation is then applied to compute the initial volume of pressurizing gas in each tank, assuming that the final volume of the gas will be equal to the total volume of the tank. Subsequently, a while loop is employed to determine the evolution of the system over time. Initially, the volumetric flow rates are computed from the mass flow rates. A time step (0.1 s) is then selected to compute the variation of volume dependent on that step.

By calculating the change in volume (ΔV) of pressurized gas, exploiting the adiabatic relation again, with the initial condition P_{tank_i} and $V_{press,tank_i}$, the new pressure inside the tanks can be obtained. Subsequently, the new chamber pressure (P_c), fuel flow velocity (u_{fuel}), and oxidizer flow velocity (u_{ox}) can be evaluated by solving the system, in which all equations can be expressed in terms of these three parameters. The following system is then incorporated:

$$\begin{cases} P_c = P_{t_{RP1}} - \Delta p_{inj} - \Delta p_{feed_{RP1}} - \Delta p_{dyn_{RP1}} \\ P_c = P_{t_{LOX}} - \Delta p_{inj} - \Delta p_{feed_{LOX}} - \Delta p_{dyn_{LOX}} \\ \dot{m} = \dot{m}_{RP1} + \dot{m}_{LOX} \end{cases} \quad (4.14)$$

Where the distributed losses of the pipelines and the concentrated losses given by the valves are included within the Δp_{feed} .

The solution of the system of non-linear equations employs the built-in Matlab function *fsolve*. The initial point for the iteration process is determined by selecting the values of the three parameters from the preceding step of the cycle.

Using the updated values of u_{fuel} and u_{ox} , the mass flow rates ($\dot{m}_{RP1}$ and $\dot{m}_{LOX}$) are computed, consequently determining the total mass flow rate ($\dot{m}$) and the new O/F ratio.

Subsequently, the new values of the mass flow rates and pressure in tanks are utilized for the computation of the new friction factor through an interpolation method. Similarly, the pressure drops due to the valve are obtained using an interpolation method with the $\dot{m}$ as input. These values are then utilized in the subsequent cycle in the system 4.14.

Following this, the new P_{ext} is computed from the P_c , maintaining a constant expansion ratio (ϵ) and specific heat ratio of the combustive gases (γ_{cg}), so the P_e/P_c ratio is expected to remain constant because the expansion ratio is constant, according to the equation:

$$\frac{1}{\epsilon} = \left(\frac{\gamma_{cg} + 1}{2} \right)^{\frac{1}{\gamma_{cg}-1}} \cdot \left(\frac{P_e}{P_c} \right)^{\frac{1}{\gamma_{cg}}} \cdot \sqrt{\frac{\gamma_{cg} + 1}{\gamma_{cg} - 1} \cdot \left[1 - \left(\frac{P_e}{P_c} \right)^{\frac{\gamma_{cg}-1}{\gamma_{cg}}} \right]} \quad (4.15)$$

All the preceding calculations are needed for the inspection of the Thrust (T), specific impulse (I_{sp}) and flow exit velocity (u_{ext}) profiles.

Upon completing all computations, the temperature of LOX is assessed to ensure it does not drop below the freezing point of the liquid. The adiabatic relation with pressures and temperatures is also employed for this purpose, with the initial value of the LOX temperature (T_{LOX}) set at 90K.

Finally, the following conditions are selected as re-entering conditions for the while loop iterations:

$$V_{He_{RP1}} < V_{tank_{RP1}} \quad V_{He_{LOX}} < V_{tank_{LOX}} \quad P_c > P_{c_{min}} \quad (4.16)$$

where $P_{c_{min}} = 20 \text{ bar}$. With this iteration methodology, it is possible to compute the time evolution of the rocket performance parameters.

Chapter 5

Results

5.1 Nominal design

5.1.1 Performance parameters

P_{exit} [bar]	z [km]	c^* [m/s]	u_{exit} [m/s]	I_s [s]	c_T [-]
0.432	25.25	1774.4	3230.1	340.18	1.8807

Table 5.1: Performance parameters

It is possible to verify that the I_s is approximately 340 s, which is the suggested maximum value for the use of blowdown architecture according to [7] in order to avoid significant pressure losses; in addition, in time profile it is decreasing, reaching an acceptable value. The c^* value is acceptable according to E.3.

5.1.2 Combustion chamber

A_{cc} [m ²]	ϕ_{cc} [m]	L_{cc} [m]	M_{cc} [-]
0.0012	0.0386	0.0982	0.0553

Table 5.2: Combustion chamber design

As opposed to the first iteration, done by imposing the Mach number at 0.2, it is achieved a more reasonable shape of the combustion chamber by setting the contraction ratio. The resulting length allows the tanks to be longer, in order to leave more space around them for the pipes and coating. The combustion chamber is rather small, but this value is expected as the required thrust is low.

5.1.3 Nozzle

The maximum performance approach represents the chosen nozzle configuration. The MPA configuration is the most efficient in terms of thrust, since it minimizes the losses. Conversely, the minimum length approach yields poorer thrust efficiency. However, it enables the minimization of the divergent section length by 40%. This leads to reduced space requirements for the divergent of the nozzle and overall weight savings. Lastly, the conical configuration offers intermediate performance compared to the previous two options, with the added advantage of a simpler geometric design. The divergence loss coefficients for the three possible configurations are reported in Table 5.3:

Nozzle configuration	λ	Loss percentage
Conical	0.9830	1.65%
RAO MLA	0.9741	2.51%
RAO MPA	0.9928	0.70%

Table 5.3: Thrust losses for different configurations

Since there is no constraint in the length of the divergent, the considered configuration is the one coming from the MPA, and the length of the single sections of the nozzle are shown in Table 5.4:

Convergent length	Divergent length	Total length
2.33 cm	16.00 cm	18.33 cm

Table 5.4: Nozzle dimensions

5.1.4 Tanks and Pressurising Gas

VR	$V_{tank,f} [m^3]$	$V_{tank,ox} [m^3]$	$\phi [m]$	$L_f [m]$	$L_{ox} [m]$	$th_{tank} [mm]$	$m_{tank,f} [m]$	$m_{tank,ox} [m]$
1.59	0.4848	0.7717	0.9621	0.6670	1.0615	1.1	1.2384	1.9710

Table 5.5: Tank design results

There is proof of the correctness of the volumetric ratio value in E.3; this value is shifted towards LOX (a higher volume of oxidizer will be discharged) but, since the oxidizer has a higher density with respect to RP-1, the difference in the volumes will be lower than the one it would have been if the two substances had identical densities: this is reflected in the tank volumes.

The diameter of the tanks is the same, as explained in 4.1.5, and it is shown that there is enough space for both the thermal coating and the pipelines. The longitudinal dimensions in the blueprint 3.1.1 show that the system fulfill the length constraint, achieving exactly 2 m. The approach already took some precautions in order to account for some of the changes usually applied in further detailed developments (i.e. more precise design of the pipes). The tank walls are thin: this is due to the choice of carbon fiber, which also allows to save a lot of mass thanks to its high tensile strength and low density, as shown by their extremely light weight. It is possible to compare them with the values that would have been found if stainless steel was selected: these values show how good the behaviour of the carbon fiber really is, as they are both below 2 kg, saving over 50 kg. Though, this is to be considered with respect to the cost of carbon fiber and its manufacturing which is significantly more expensive and difficult to manufacture. A further analysis could be carried out by calculating the effects of the difference in mass in terms of how much it impacts the fuel consumption and, consequentially, how much it affects the whole mission, keeping in mind the requirements. Moreover, the system appears to be best suitable for an upper stage, thus the mass saving is crucial, since during the takeoff it will act as a payload.

It is also important to note that the effective volumes occupied by RP-1 and LOX will be $0.1643 m^3$ and $0.2558 m^3$ respectively, showing the amount of percentage in the tank volumes occupied by the pressurising gas. This could force a thought about reconsidering the system architecture, as the volumes are not ideally exploited.

5.1.5 Injection plate

$A_{tot,fuel} [m^2]$	$N_{inj,fuel}$	$\phi_{fuel} [mm]$	$A_{tot,ox} [m^2]$	$N_{inj,ox}$	$\phi_{ox} [mm]$
$3.3915 \cdot 10^{-6}$	18	0.500	$6.4036 \cdot 10^{-6}$	18	0.673

Table 5.6: Injection plate design results 1

As explained in 4.1.6, a doublet configuration is selected, so it is imposed that the injectors for the two fluids are the same number. This of course will result in a larger diameter for the oxidizer injectors as a higher mass flow rate needs to be discharged. The diameter of the fuel injector is rather small and the one for the oxidizer injectors, which is computed, is just narrowly higher. This is a good factor regarding the atomization of the fluids, and it is still an acceptable dimension for what regards the additive manufacturing capabilities.

It is possible to modify the number of injectors by rising or lowering the diameter of the RP-1 injectors: this could be interesting in order to reach a prime number on the circumference of the injection plate, such that some instabilities and resonances are avoided. The precise placement of the holes on the injection plates is left to a further development stage.

$u_{inj,fuel} [m/s]$	$u_{inj,ox} [m/s]$	$\delta_{fuel} [deg]$	$\delta_{ox} [deg]$	$\xi [deg]$
34.7833	29.3198	30	36.3824	66.3824

Table 5.7: Injection plate design results 2

The exit velocities are reasonable, as they are in a range from 10 – 50 m/s, and the fuel one is higher as, considering the same discharge coefficient and the same injection ΔP for both fluids, they are dependent only

on the inverse squared of the densities.

Regarding the injection angles it is important that the impinging angle is close to 60° , as stated in 4.1.6, to obtain a good mixing of the flows.

5.1.6 Cooling system

It is verified that cooling the combustion chamber through the cooling jacket can be done, because the high heat flux (q) can be dissipated. A highly performing material is chosen to be capable of withstanding high temperatures, with a low thermal conductivity coefficient (k_{cond}) to minimize heat transfer through conduction. Usually it is recommended to use higher k_{cond} to reduce the wall temperature drop, but in the case analyzed, using the same thickness, it would increase the heat flux through the wall. This result permits to obtain a very small minimum combustion chamber thickness of 0.0048 m. With the calculated temperature variation of 270 K, computed through D.17, it results that the final temperature reached by RP1 is approximately 570 K, lower than its saturation and coking temperature. This enables the fuel to effectively cool the combustion chamber through the establishment of a cooling jacket through the method explained in 4.1.7. The final geometry for the channel can be summarized as in the table 5.8, according to the model D.2 in which the coolant flows through rectangular slots, or channels, machined into the chamber liner. The machined channels are closed out and the liner is enclosed in a structural support shell.

N_{ch}	$d_{1,ch}[m]$	$d_{2,ch}[m]$	$t_{ch}[m]$
22	$0.5 \cdot 10^{-3}$	$0.5 \cdot 10^{-3}$	$5 \cdot 10^{-3}$

Table 5.8: Design of cooling channel

It is noticeable from the overall analysis, as indicated by 5.9, that the highest heat flux occurs in the throat section. Moreover, in this section, it's not feasible to employ the cooling jacket. Therefore, it's proposed to utilize the dumb cooling method along with the combined effect of an insulating ceramic material. In the dumb cooling a small amount of fuel is diverted from the main line, passed through cooling passages and ejected. The mechanism is similar to that of regenerative cooling. The coolant, however, becomes superheated as it flows toward the nozzle exit, where it is expanded overboard at reasonably high temperatures and velocities, thus contributing to some thrust. [41], D.1 Zirconia Ytria is the suggested ceramic material to use in the nozzle section because it is suited to resist up to 2500 K. Moreover, its remarkable heat resistance makes it the best insulation material currently available [42]. In the design illustrated in D.3, zirconia tubes with a diameter of 1/8 inch are suggested. These tubes are filled with a foam made of the same zirconia material, creating a robust insulation system. This innovative approach represents the cutting edge of thermal protection technology, offering superior performance in demanding environments. The manufacturing of zirconia tubes can be accomplished using conventional powder metallurgy techniques or slip casting methods. [42]

	Combustion chamber	Throat	Nozzle exit
M	0.0553	1	4.9670
Re	$9.5966 \cdot 10^4$	$3.2701 \cdot 10^5$	$1.0064 \cdot 10^5$
Pr	0.6216	0.6268	0.0661
Nu	222.3299	594.3438	117.9203
$h_{cg} [W/(m^2K)]$	$2.0692 \cdot 10^3$	$1.7054 \cdot 10^4$	$1.1226 \cdot 10^3$
$k_{cg} [W/(m^2K)]$	0.3593	0.3394	0.9625
$\mu_{cg} [Pa \cdot s]$	$1.0639 \cdot 10^{-4}$	$1.0192 \cdot 10^{-4}$	$3.8741 \cdot 10^{-5}$
$q [W/(m^2K)]$	$4.0810 \cdot 10^6$	$3.2673 \cdot 10^7$	$4.2784 \cdot 10^5$
$\rho_{cg} [kg/m^3]$	3.6882	2.3001	0.0119

Table 5.9: Results of cooling design

5.2 Time profile

This paragraph reports the results obtained in the time profile design of the different quantities involved in the analysis. The *while* loop described in section 4.2 ends when the oxidizer tank is full of helium, hence when all the oxidizer on-board has left its vessel. The burning time reads $t_b = 2030.2$ s, which is around 34 minutes. This value is justified by the fact that the required thrust is rather low, thus demanding a limited mass flow rate. The dimensions allocated to the propulsion system allow to load an amount of propellant capable of ensuring a burning time of around half an hour. In figure 5.1 the combustion chamber pressure time profile is reported. P_c starts from the initial value of 50 bar and decreases until all the oxidizer is consumed. The lower pressure limit of 20 bar is not reached, since the final pressure chamber is of 25.68 bar. Hence, the engine combustion chamber consistently operates within its designated working limits.

Pressure time evolution in the two tanks is presented in figure 5.2. It is clear that the initial pressure in the oxidizer tank exceeds the one of the fuel tank. As the propellants are gradually expelled from the vessels, the pressurizing gas expands, leading to a decrease in pressure. The oxidizer tank pressure decreases more compared to the fuel tank pressure, indicating a greater expulsion of oxidizer relative to fuel.

As a result, the oxidizer mass flow rate decreases at a slower rate than the fuel mass flow rate, leading to a rise in the oxidizer-to-fuel ratio. Since the thermodynamic process in the tanks is described as an adiabatic process, the temperature decreases from its initial value in both tanks. Particularly important is the temperature in oxidizer tank, as LOX is a cryogenic propellant and therefore needs to be maintained above its freezing point. Figure 5.3 illustrates the variation of this temperature over time. It can be seen that the oxygen freezing temperature of 55 K [7] is never reached (final temperature of around 69 K), avoiding the risk of freezing.

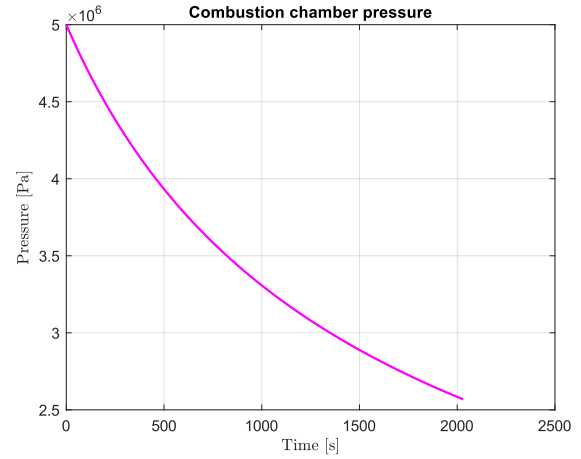


Figure 5.1: Combustion chamber pressure evolution

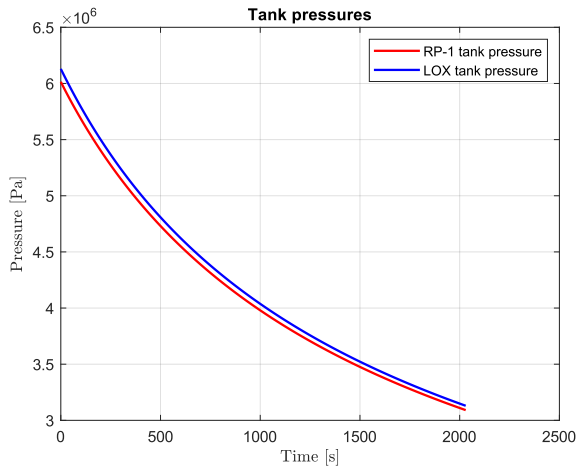


Figure 5.2: Tanks pressure evolution

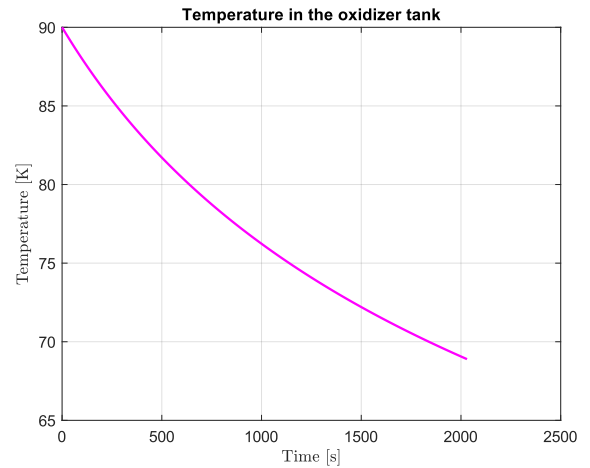


Figure 5.3: Temperature in Oxidizer tank

The time evolution of mass flow rate of the two propellant is shown in Figure 5.4, together with the total mass flow rate. The latter decreases as a consequence of the pressure chamber's diminishing pressure.

It is interesting to highlight the behaviour of the O/F ratio, which is not constant in the blowdown system since the mass flow rates of oxidizer and fuel decrease differently, as stated above. The O/F ratio initially increases from a starting value of 2.24 until reaching a peak, after which it stabilizes at a particular value. This is shown in Figure 5.5.

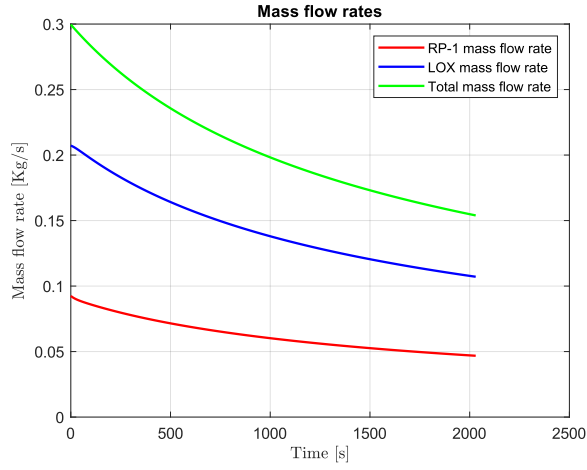


Figure 5.4: Mass flow rate evolution

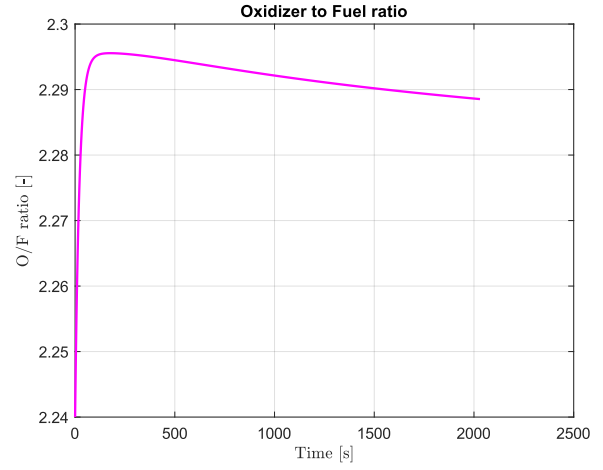


Figure 5.5: O/F ratio evolution

5.2.1 Thrust, Specific Impulse and Total Impulse

It is essential to plot the time evolution of the performance parameters. Particularly significant is the behavior of thrust, which diminishes over time from its initial value of 1 kN. This decline is attributed to the decrease in both mass flow rate and exit pressure, resulting in reductions in both the dynamic and static contributions to the thrust equation. Figure 5.6 shows the time evolution of these terms, together with the thrust profile during the whole burning time. As expected, the dynamic term is prevalent and its reduction is evident, while the static contribution is almost constant. Static term is lower because the expansion ratio is high, so the flow is expanded more. The presence of this variable static contribution also affects the behaviour of the specific impulse, and, hence, of the thrust coefficient. Indeed, the specific impulse depends on both thrust and mass flow rate: since the static term is not constant, also I_{sp} changes over time.

However, this change is very small because static term is almost negligible compared to the dynamic one. Consequently, the specific impulse evolution illustrated in Figure 5.7 appears to be constant, therefore in the graph it is not possible to appreciate the real trend.

Another crucial parameter to calculate is the *total impulse*, defined as the thrust integrated over the burning time. The total impulse is proportional to the total energy released by or into all the propellant utilized by the propulsion system. In Table 5.10 are reported the values of I_{tot} for the nominal thrust and for the maximum performance approach of the nozzle. When considering nozzle losses, thrust decreases from the nominal value, leading to a reduction in the total impulse as well.

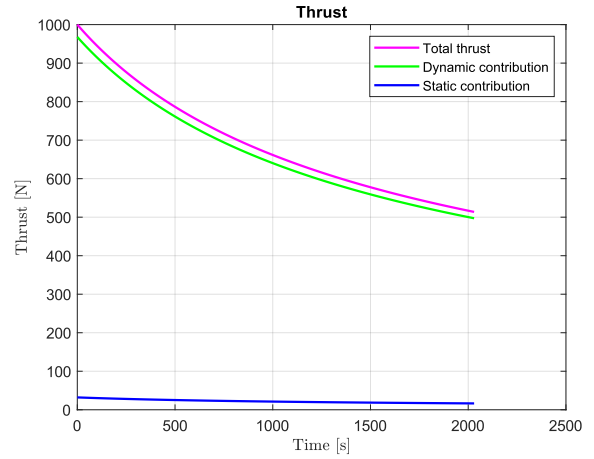


Figure 5.6: Thrust evolution

	Nominal	MPA
Total impulse [$N \cdot s$]	$1.3977 \cdot 10^6$	$1.3880 \cdot 10^6$

Table 5.10: Total impulse

Thrust profile in the parabolic nozzle differs from the nominal one, since losses cause a decrease in the dynamic contribution of thrust. In Figure 5.8 is shown the slight difference between the two thrust profiles: the choice of MPA approach for the designed RAO nozzle is also graphically justified, as the thrust using the MPA is only lightly below the ideal one.

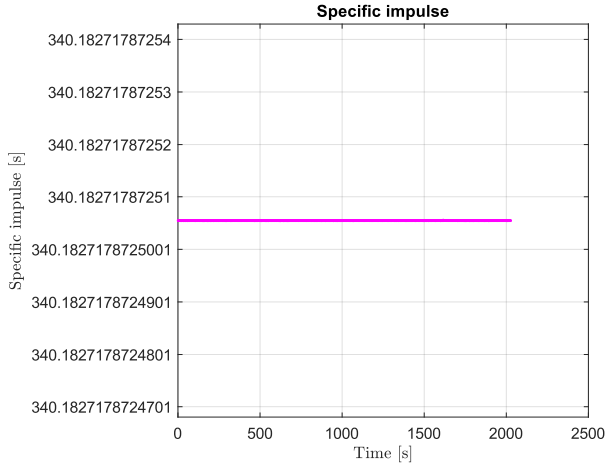


Figure 5.7: Specific impulse evolution

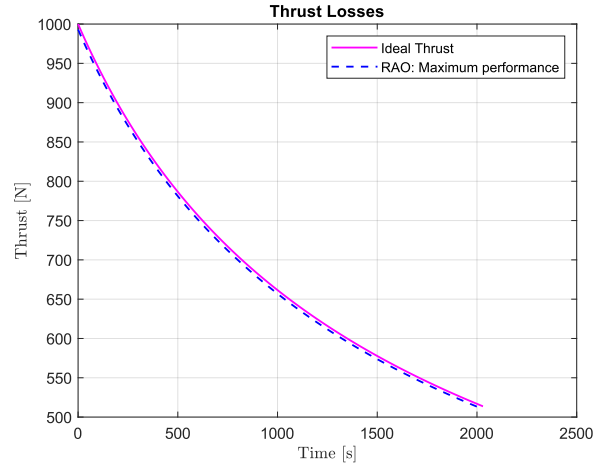


Figure 5.8: Thrust losses

5.3 Effects of AM uncertainties

Introducing uncertainties in the hole diameters of the injection plate can yield different results from the nominal performance of the engine. These effects are shown in the following figures. Different types of uncertainties have been considered. While *feature benchmark LB* (lower boundary, meaning the case with the tightest holes) *B.1.4* and *UP* (upper boundary, in which the diameters are the largest) are purely due to geometrical imperfections leading to a reduction or increment of the diameter of the orifices [1], the *experimental deviations LBB.1.3* and *UBB.1.2* are retrieved from the paper [13]. *Gloria et al.* after an experimental campaign computed a variation of the $C_d A$ [mm^2], from the expected one that may also depend on factors other than the geometrical imperfections, but could not be understood by the experimenter. It is clear that the thrust profile differs from the nominal one: increasing the hole diameter results in higher thrust while, decreasing the diameter, a lower thrust is obtained. This is strictly related to the mass flow rate evolution over time of both RP-1 and LOX, represented in Figures 5.9. With larger holes, a greater mass flow passes through, leading to a reduction in total burning time (as propellant is consumed faster). Conversely, smaller diameters result in a lower mass flow rate, hence extending the burning time. The range of burning times, accounting for existing uncertainties, is documented in Table 5.11:

	Lowest	Nominal	Highest
Burning time [min]	27.1967	33.8367	39.7833

Table 5.11: Burning time considering injection plate uncertainties

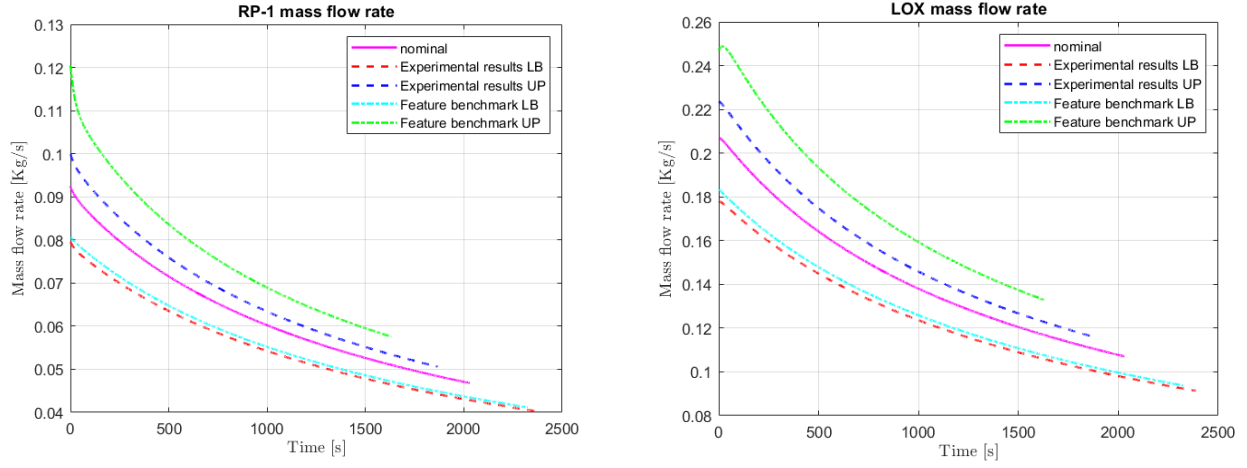


Figure 5.9: Mass flow rates due to uncertainties

In Figure 5.10 is shown the thrust evolution considering the various type of injection uncertainties. As expected, as the section area diminishes the overall thrust lowers whilst by increasing the diameter of the injectors the Thrust remain higher but has a more descending trend.

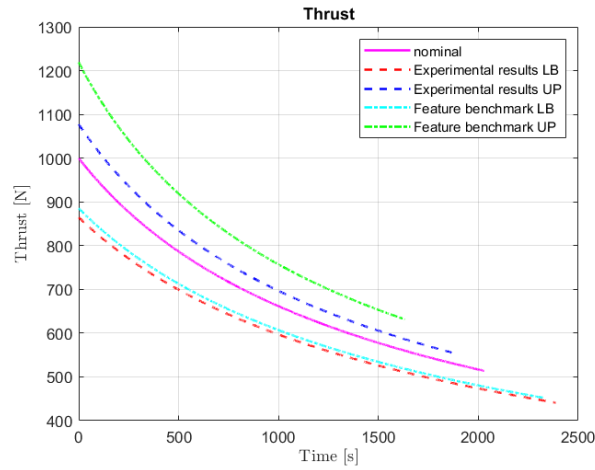


Figure 5.10: Thrust evolution with injection uncertainties

Chapter 6

Conclusions

The analysis and design undertaken has been performed with method and paying careful attention to the assumptions made, to the validity of the results obtained, to the general feasibility of the system and to other important aspects, such as the material, valves and other components selection.

Another crucial aspect is the propellant couple behaviour and properties: RP-1/LOX is a semi-cryogenic couple and a thorough analysis on the variability of LOX temperature has to be taken into account, as done in this project thanks to the study of an insulation coating made of PU foam and materials capable to withstand such a low temperature. Regarding the thermal analysis in the combustion chamber, the design of a cooling jacket is proposed, showing the feasibility of the dissipation of the required heat flux while maintaining RP-1 below the saturation temperature.

In the modeling approach, a strong correlation between many parameters in the blowdown system is evident: in particular, the fact that the pressure in the two tanks decreases in time in different ways, and this is related to a number of factors, such as the densities of the fluids, their velocities in the pipelines, the combustion chamber pressure and the O/F ratio.

The design of the tanks, and of the system in general, is strongly affected by the choice of a blowdown architecture. This entails advantages, such as the simplicity of the system owed to the lack of additional tanks and of a pressure regulation system, as well as drawbacks, such as the fact that a high percentage of the tanks volume is occupied by the pressurising gas. A further analysis that can be carried out related to this matter is the difference in fuel and oxidizer volume that can be stored with respect to a pressure regulated or a re-pressurized blowdown system, which would require an additional tank but would allow the full capacity of the RP-1 and LOX tanks to be occupied by these fluids.

The following steps to be undertaken involve the need to relax some assumptions made, especially the ones regarding constant chamber temperature and other combustion parameters, such as molar mass and specific heat ratio of the mixture in order to obtain a more precise simulation. Furthermore, a more detailed analysis on the system components is to be performed, e.g. the injection plate holes configuration has to be studied more accurately in order to guarantee a better mixing and less pressure losses. Further developments shall be carried out to assess the entity of hydraulic shocks when performing an emergency shut-off of the motor. Moreover a different architecture may be considered, exploiting a wider LOX tank connecting the piping at the periphery of vessel and consequently removing the necessity of an interspace between the propellants.

Appendix A

Nozzle losses

A.1 Boundary layer losses

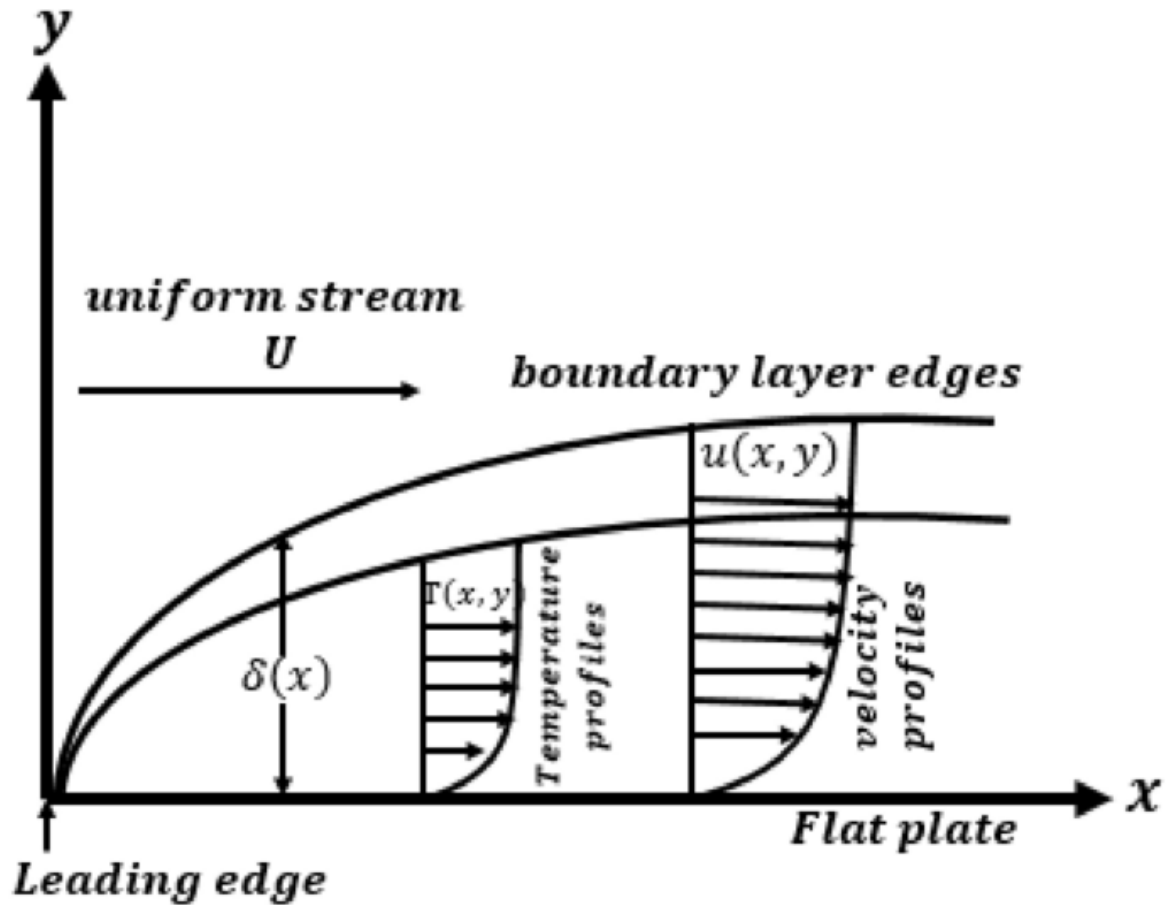


Figure A.1: Velocity and temperature profile in boundary layer

A.2 Divergence losses

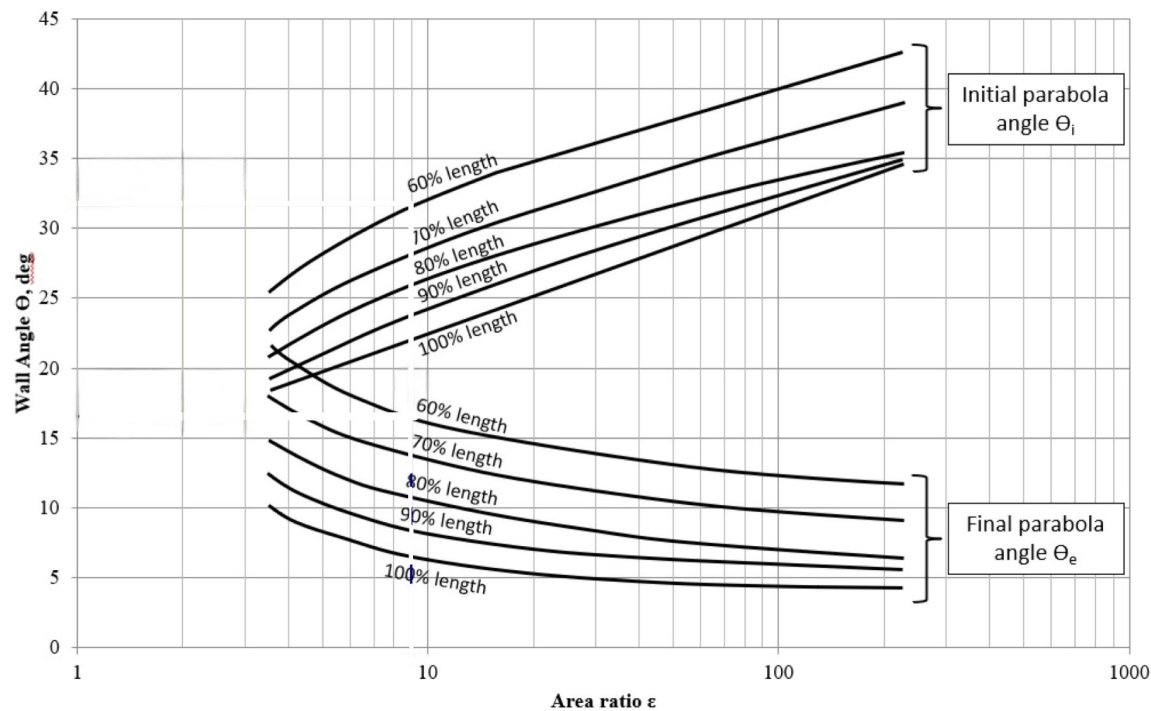


Figure A.2: RAO curves

Appendix B

Additive manufacturing

B.1 Definitions

B.1.1 Cilindricity:

describes the conformity to a perfect cylinder. It describes the distance of two concentric cylinders in which all points of the surface of the cylindrical feature fall.[1]

B.1.2 Experimental Upper Bound (UB):

maximum value of positive deviation of C_dA for the oxidizer line obtained in the experimental campaign. [13]

B.1.3 Experimental Lower Bound (LB):

maximum value of negative deviation of C_dA for the fuel line obtained in the experimental campaign. [13]

B.1.4 Geometrical Lower and (LB) Upper Bound (UB):

From [1] the LPBF fall in the *fine class* of tolerances. This value is applied to the injection plate orifices to compute maximum and minimum diameters of those.

B.2 Figures

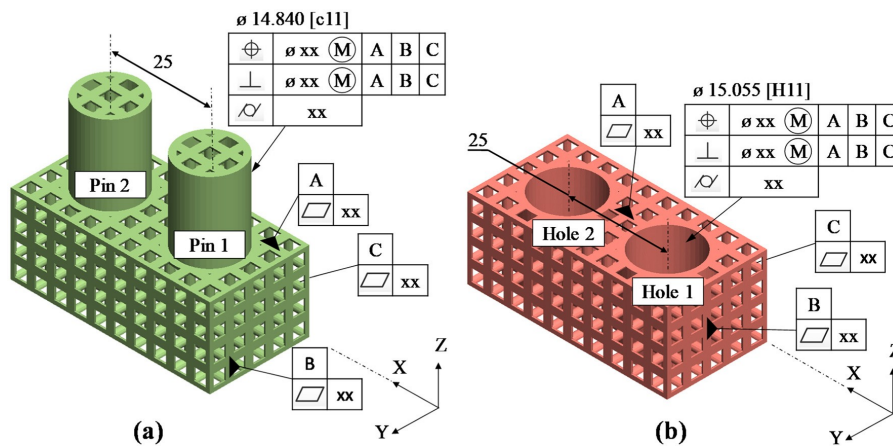


Figure B.1: Benchmark piece designed to test uncertainties on assemblies components

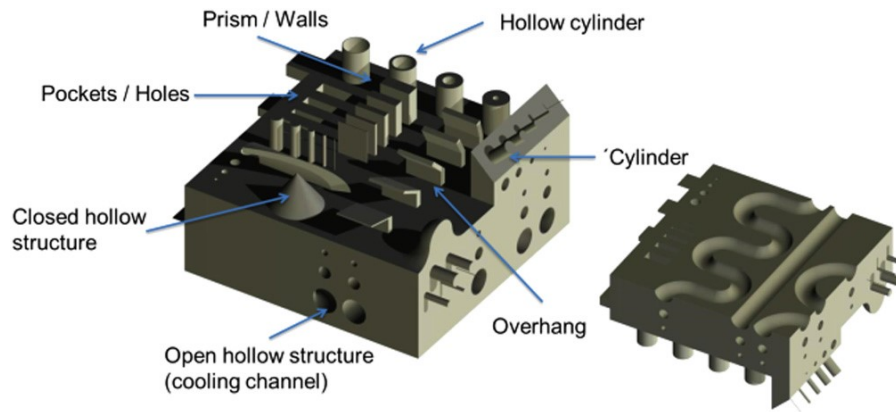


Figure B.2: Benchmark piece designed to test uncertainties on inner features [1]

Process	IT-Classes (DIN EN ISO 286-1)											
	5	6	7	8	9	10	11	12	13	14	15	16
Casting												
Sintering												
Drop forging												
Precision forging												
Cold extrusion												
Milling												
Cutting												
Turning												
Drilling												
Face milling												
Planing												
Stripping												
Circular grinding												
Additive manufact.												
FDM												
LS												
LM												

Figure B.3: Comparison of the results with commonly used techniques in terms of ISO tolerances [3]

Appendix C

Blowdown architecture

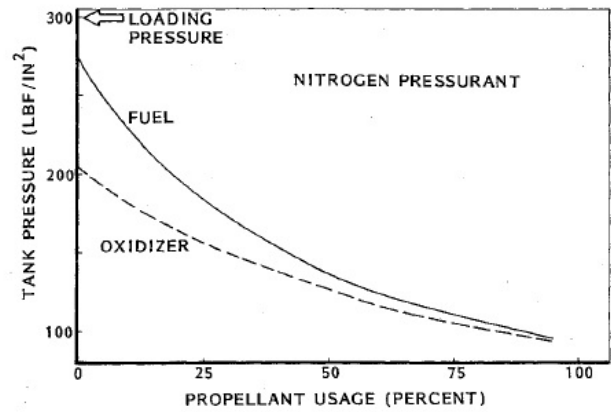


Fig. 1 Tank pressure blowdown using nitrogen.

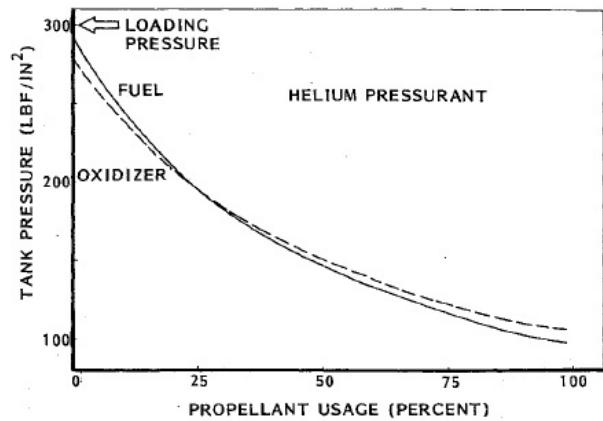


Fig. 2 Tank pressure blowdown using helium.

Figure C.1: Tank pressure blowdown using nitrogen and helium

Appendix D

Modelling Approach

D.0.1 Equations

Adiabatic relations

$$T_{He,fin,RP1} = T_{He,in,RP1} \left(\frac{P_{He,fin,RP1}}{P_{He,in,RP1}} \right)^{\frac{\gamma_{He}-1}{\gamma_{He}}} \quad (D.1)$$

$$T_{He,fin,LOX} = T_{He,in,LOX} \left(\frac{P_{He,fin,LOX}}{P_{He,in,LOX}} \right)^{\frac{\gamma_{He}-1}{\gamma_{He}}} \quad (D.2)$$

$$V_{He,in,RP1} = V_{He,fin,RP1} \left(\frac{P_{He,fin,RP1}}{P_{He,in,RP1}} \right)^{\frac{1}{\gamma_{He}}} \quad (D.3)$$

$$V_{He,in,LOX} = V_{He,fin,LOX} \left(\frac{P_{He,fin,LOX}}{P_{He,in,LOX}} \right)^{\frac{1}{\gamma_{He}}} \quad (D.4)$$

Performance parameters

$$c^* = \sqrt{\frac{(R/M_{mol}) \cdot T_c}{\gamma \cdot \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{\gamma-1}}}} \quad (D.5)$$

$$u_{ext} = \sqrt{2 \cdot \frac{\gamma}{\gamma-1} \cdot \frac{R}{M_{mol}} \cdot T_c \cdot \left[1 - \left(\frac{P_e}{P_c} \right)^{\frac{\gamma-1}{\gamma}} \right]} \quad (D.6)$$

$$I_{sp} = \frac{T}{\dot{m} \cdot g_0} \quad (D.7)$$

$$c_T = \sqrt{\frac{2 \cdot \gamma^2}{\gamma-1} \cdot \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{\gamma-1}} \cdot \left[1 - \left(\frac{P_e}{P_c} \right)^{\frac{\gamma-1}{\gamma}} \right]} + \varepsilon \cdot \frac{P_e}{P_c} \quad (D.8)$$

$$A_t = \frac{T}{I_{sp} \cdot g_0} \quad (D.9)$$

Cooling

$$P_r = \frac{\mu_{cg} \cdot c_p}{k_{cg}} \quad (D.10)$$

$$Nu = 0.0265 \cdot Re_e^{0.8} \cdot Pr^{0.3} \quad (D.11)$$

$$h_{cg} = \frac{Nu \cdot k_{cg}}{\phi} \quad (D.12)$$

$$r = \frac{1 + Pr^{1/3} \cdot \frac{\gamma-1}{2} \cdot M^2}{1 + \frac{\gamma-1}{2} \cdot M^2} \quad (D.13)$$

$$T_{aw} = r \cdot T_0 \quad (D.14)$$

$$q = h_{cg} \cdot (T_{aw} - T_{material,hot}) \quad (D.15)$$

$$Q = q \cdot A_l \quad (D.16)$$

$$\Delta T = \frac{Q}{\dot{m}_{fuel} \cdot C p_{fuel}} \quad (D.17)$$

$$\Delta th_{wall} = \frac{K_{mat}}{q} \cdot (T_{material,hot} - T_{wall}) \quad (D.18)$$

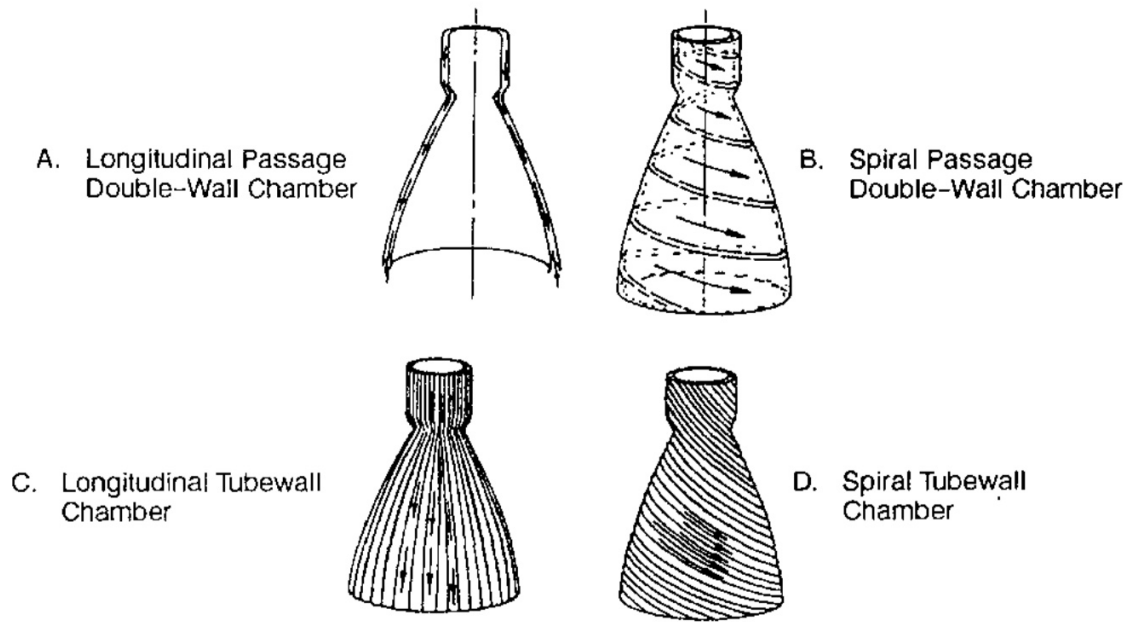


Figure D.1: Typical dumb cooled chamber fabrication methods

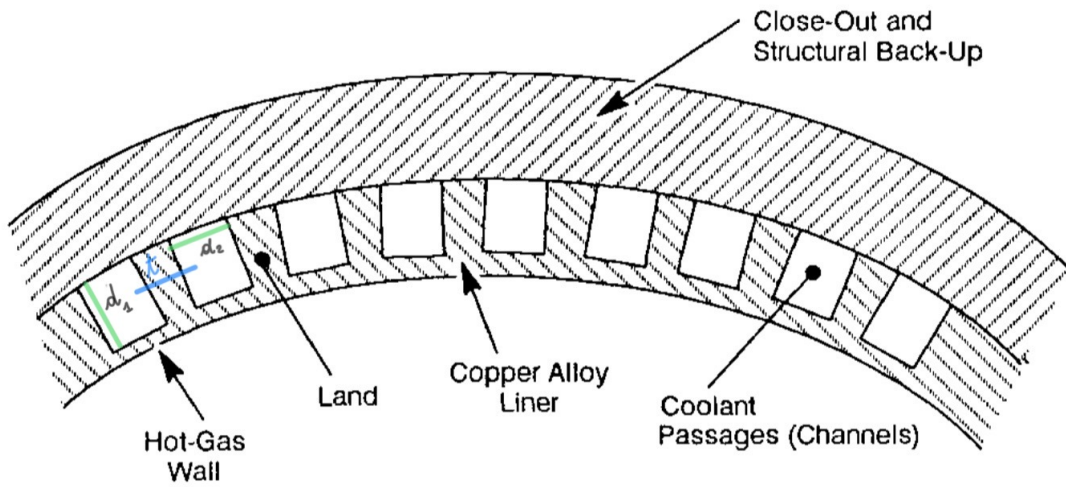


Figure D.2: Typical channel wall configuration

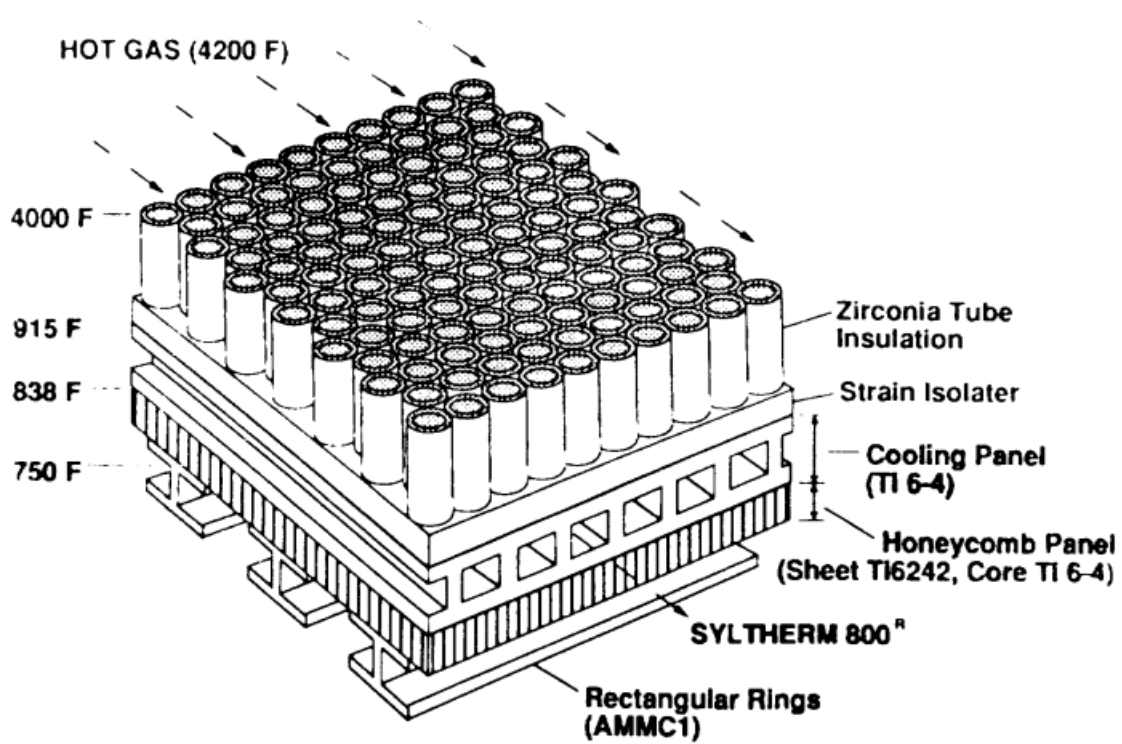
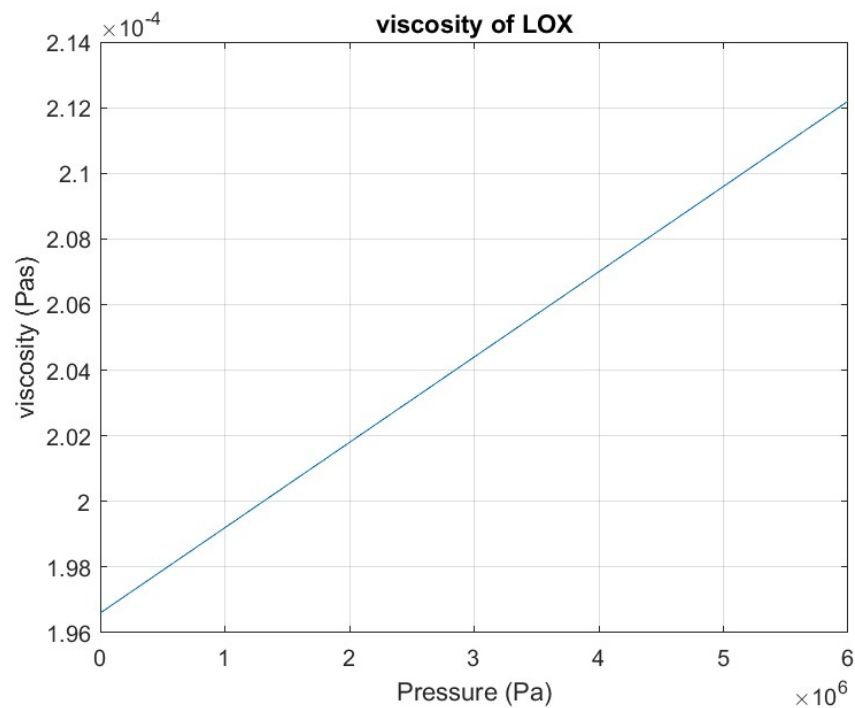


Figure D.3: Zirconia - Yttria

Appendix E

Fitting of Viscosity Curves



E.1 Viscosity of RP1

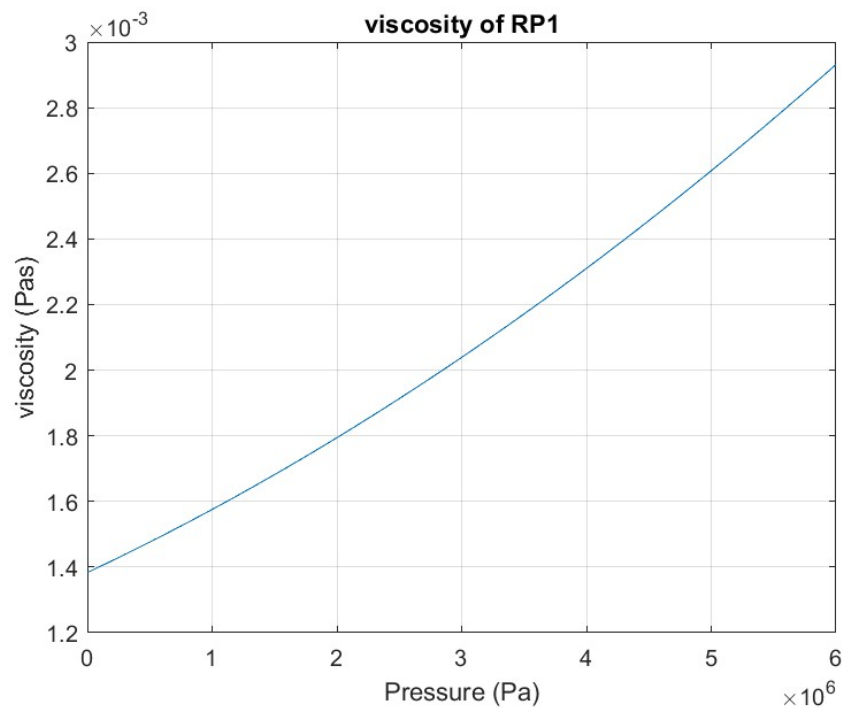


Figure E.1: Enter Caption

E.2 Moody diagram

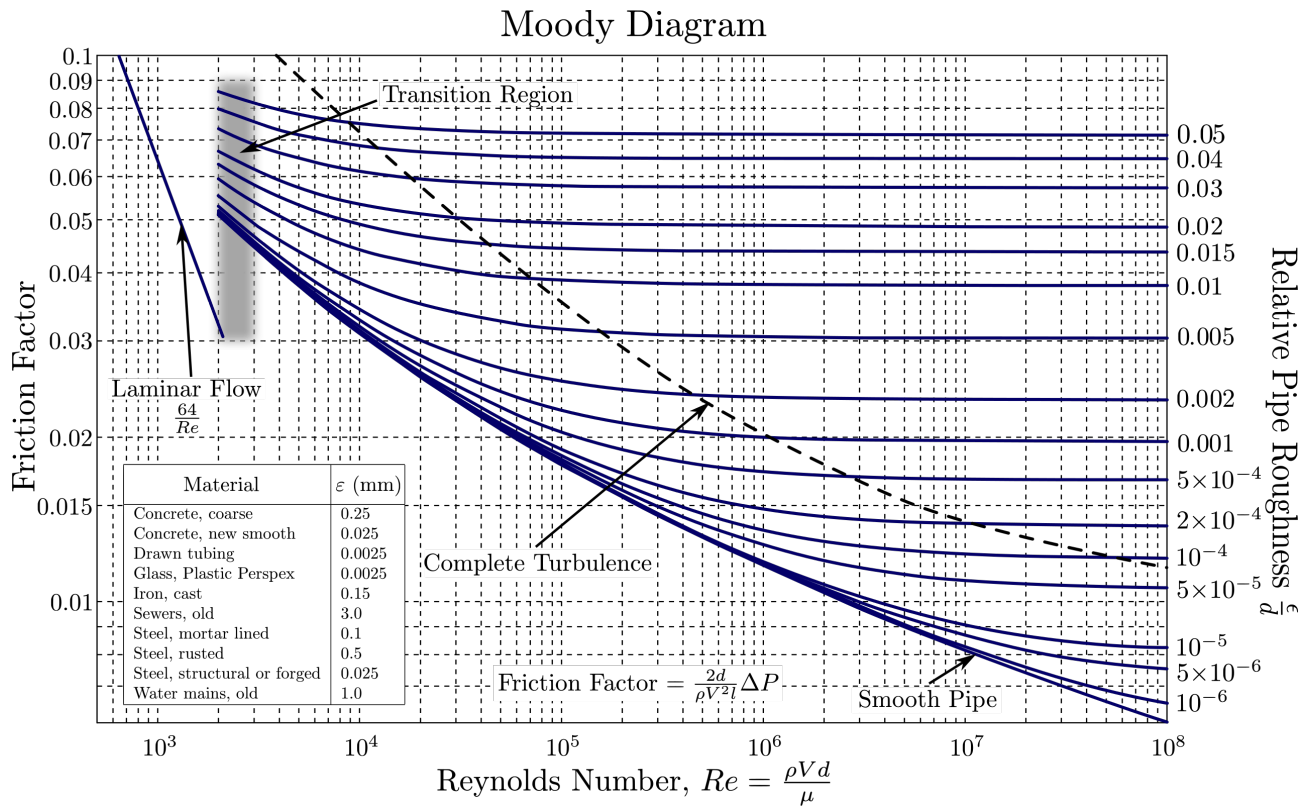


Figure E.2: Moody diagram

TABLE 5–5. Theoretical Chamber Performance of Liquid Rocket Propellant Combinations

Oxidizer	Fuel	Mixture Ratio		Average Specific Gravity	Chamber Temp.(K)	Chamber c^* (m/sec)	$\mathfrak{M}$, (kg/mol)	I_s (sec)		
		By Mass	By Volume					Shifting	Frozen	k
Oxygen	Methane	3.20	1.19	0.81	3526	1835	20.3		296	1.20
		3.00	1.11	0.80	3526	1853		311		
	Hydrazine	0.74	0.66	1.06	3285	1871	18.3		301	1.25
		0.90	0.80	1.07	3404	1892	19.3	313		
	Hydrogen	3.40	0.21	0.26	2959	2428	8.9		386	1.26
		4.02	0.25	0.28	2999	2432	10.0	389.5		
	RP-1	2.24	1.59	1.01	3571	1774	21.9	300	285.4	1.24
		2.56	1.82	1.02	3677	1800	23.3			
Fluorine	UDMH	1.39	0.96	0.96	3542	1835	19.8		295	1.25
		1.65	1.14	0.98	3594	1864	21.3	310		
	Hydrazine	1.83	1.22	1.29	4553	2128	18.5	334		1.33
		2.30	1.54	1.31	4713	2208	19.4		365	
	Hydrogen	4.54	0.21	0.33	3080	2534	8.9		389	1.33
		7.60	0.35	0.45	3900	2549	11.8	410		
	Nitrogen tetroxide	1.08	0.75	1.20	3258	1765	19.5		283	1.26
		1.34	0.93	1.22	3152	1782	20.9	292		
Red fuming nitric acid	50% UDMH	1.62	1.01	1.18	3242	1652	21.0		278	1.24
		2.00	1.24	1.21	3372	1711	22.6	289		
	50% hydrazine	3.4	1.05	1.23	3290		24.1		297	1.23
		2.15	1.30	1.20	3396	1747	22.3	289		
	RP-1	1.65	1.00	1.16	3200	1591	21.7		278	1.23
		4.1	2.12	1.35	3175	1594	24.6		258	1.22
	50% UDMH	4.8	2.48	1.33	3230	1609	25.8	269		
		1.73	1.00	1.23	2997	1682	20.6		272	1.22
Hydrogen peroxide (90%)	50% hydrazine	2.20	1.26	1.27	3172	1701	22.4	279		
		RP-1	7.0	4.01	1.29	2760	21.7		297	1.19

Figure E.3: Table containing values for different propellant couples

Appendix F

Architecture

F.1 Thickness Assessment: ASME UG-16 Eq. 1-4.1

$$\text{Factor Head } K = \frac{1}{6} \left[2 + \left(\frac{\phi_{tank}}{2h_{head}} \right)^2 \right] \quad (\text{F.1})$$

$$\text{Head Thickness } th_{head} = \frac{P_{burst} \phi_{tank} K}{2\sigma_{ult} E - 0.2P_{burst}} \quad (\text{F.2})$$

F.2 Thickness Assessment: Cylindrical approximation

$$th_{head} = \frac{P_{burst} \phi_{tank}}{\sigma_{ultimate}} [7] \quad (\text{F.3})$$

F.3 Architecture Overview

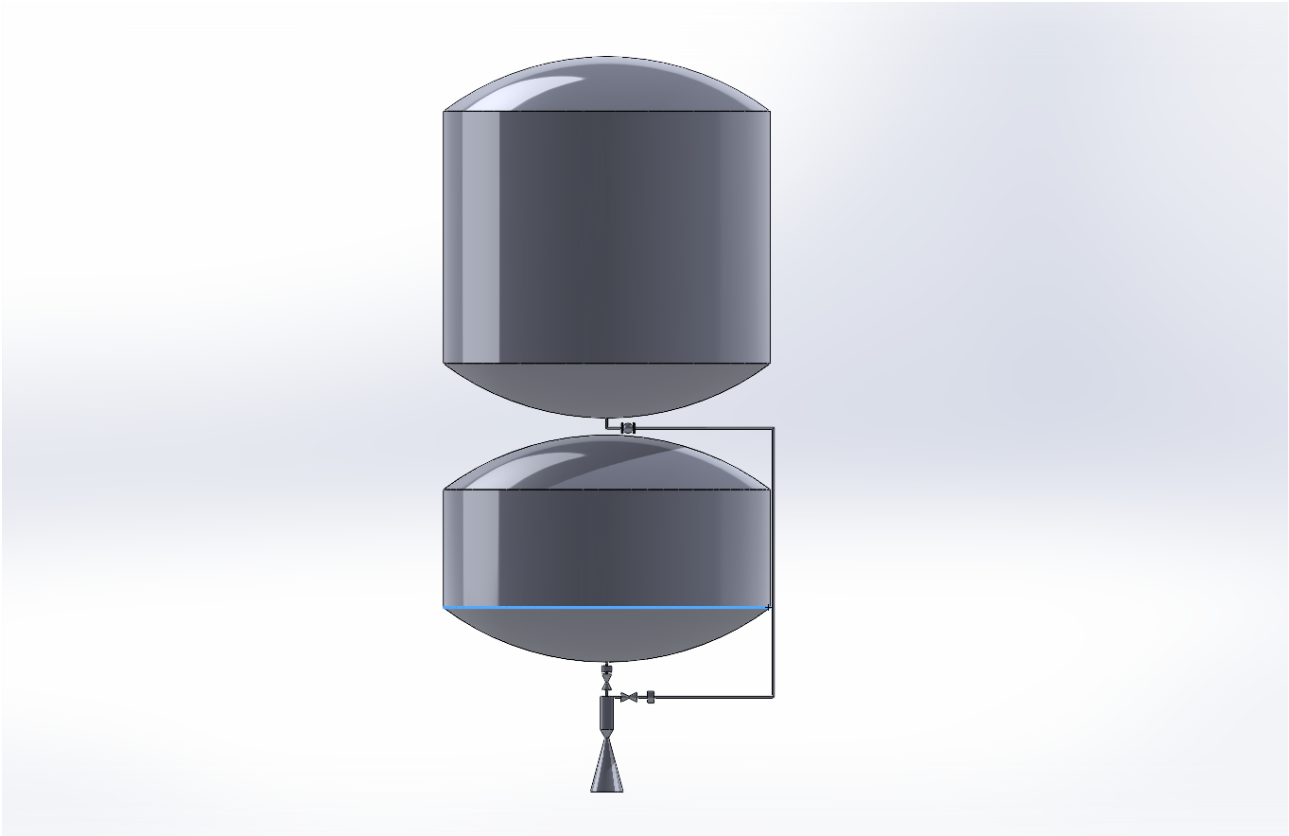


Figure F.1: 3D Depiction of the motor

F.4 Igniter

$$\text{Igniter mass } m_{igniter} = 0.12 \cdot (V_{chamber})^{0.7} \quad (\text{F.4})$$

F.5 Diaphragm

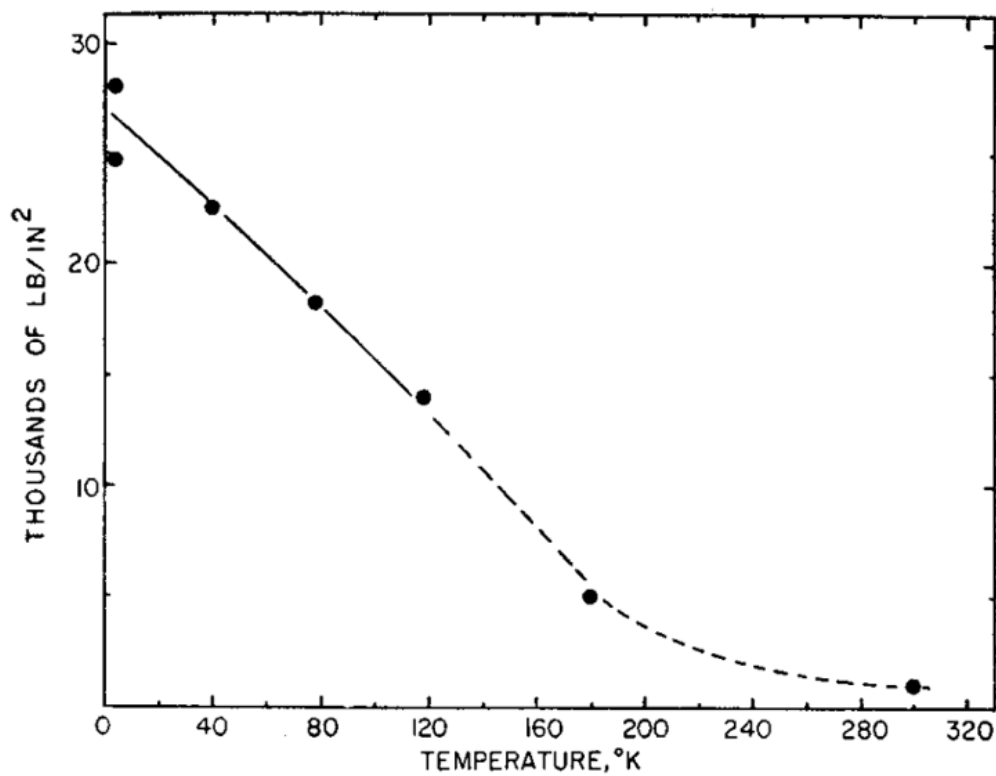


Figure F.2: Teflon ultimate tensile strength vs temperature

F.6 Insulator materials

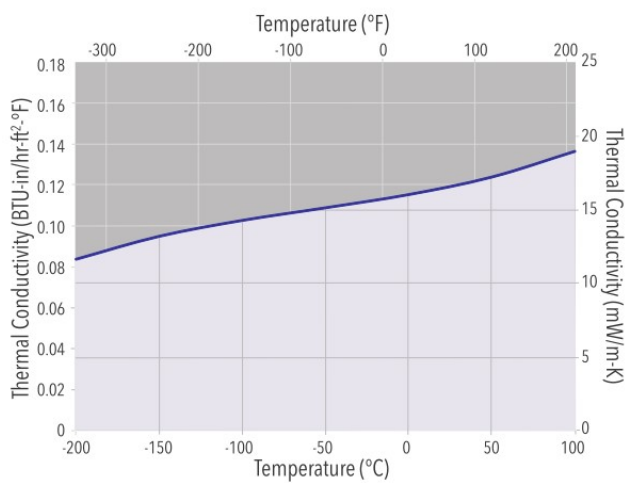


Figure F.3: Cryogel thermal conductivity

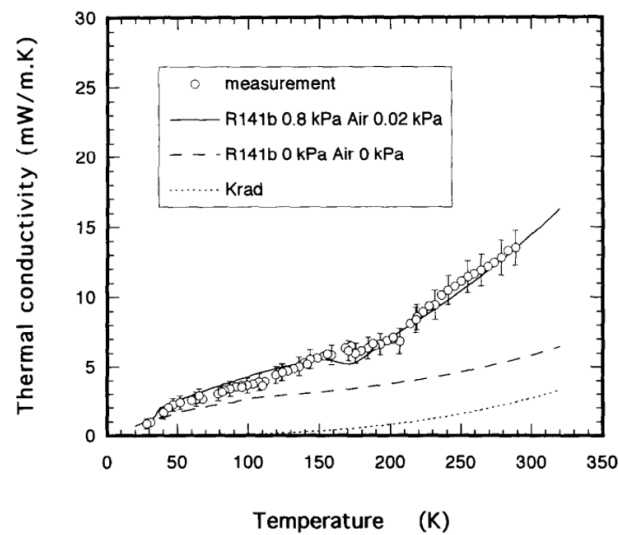


Figure F.4: Poliurethane thermal conductivity

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Declaration of Autorship

Grimaldi Samuele	<i>Modeling and programming of nominal design(focus on tanks, pressurising gas and injection plate); Partial contribution to the programming of the cooling system design.</i>
Laghi Davide	<i>Partial contribution to blowdown architecture; Pressure cascade modeling; Modeling and programming of the time profile; Programming of injection plate uncertainties changes.</i>
Pratola Oscar	<i>Literature Review: Additive Manufacturing; Piping and Valves selection, Material Compatibility: tanks, pipes and valves; Material Choice; Strength Assessments of the tanks.</i>
Rizzetto Lorenzo	<i>Material compatibility: tanks, diaphragm and insulator; Igniter: selection and sizing; Literature review: tanks disposition and external rocket case temperature; Insulator design and choice.</i>
Spinelli Giulia Leda	<i>Introduction and overview of the problem; Empty space evaluation; Part of pressure cascade regarding the distributed pressure losses; Nominal design and programming focus on: Performance parameters- Combustion Chamber-Cooling system and the relative results.</i>
Volontè Filippo Andrea	<i>Literature analysis about nozzle losses and blowdown architecture; Nozzle design; Modeling of time profile.</i>
